

# **Resolved Imaging of Globular Clusters in Andromeda**

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**Research School of Astronomy and Astrophysics**

Yi Shuen Lee

ylee9@student.unimelb.edu.au

Supervised by Dr. Dougal Mackey

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# Introduction

Star clusters are groups of stellar bodies, usually of similar age, held together by gravitational forces of the respective members within the cluster. Globular clusters are specifically a densely packed group of stars with star counts ranging from hundreds of thousands to millions. These clusters can be found in the galactic disc or the stellar halo encapsulating the galaxy. Globular clusters are usually much older than Open clusters. In fact, they are one of the oldest objects known in the universe and are therefore useful for the study of galactic archaeology. The composition of these star clusters can possibly give us some hints to the history of stellar evolution in the hosting galaxy or around the specific region in which the cluster lies.

This research project focusses on the globular star clusters of our neighbouring spiral galaxy, M31, otherwise known as the Andromeda Galaxy. The first part of the project involves a detailed analysis on two different globular clusters: PAndAS-2 (PA02) and Mayall II (NGC-224-G1 or G1) imaged by a spaced telescope - the Hubble Space Telescope (HST). The main aim in this section is to determine the cluster metallicity and distance to the cluster using the stellar-isochrone-fitting technique, as well as the procedures for running DOLPHOT – a stellar photometry package developed by Andrew Dolphin. Investigations on these globular clusters of M31 will mainly be carried out based on colour-magnitude diagrams (CMDs) as well as sharpness against magnitude, crowding against magnitude and stellar cluster radial plots.

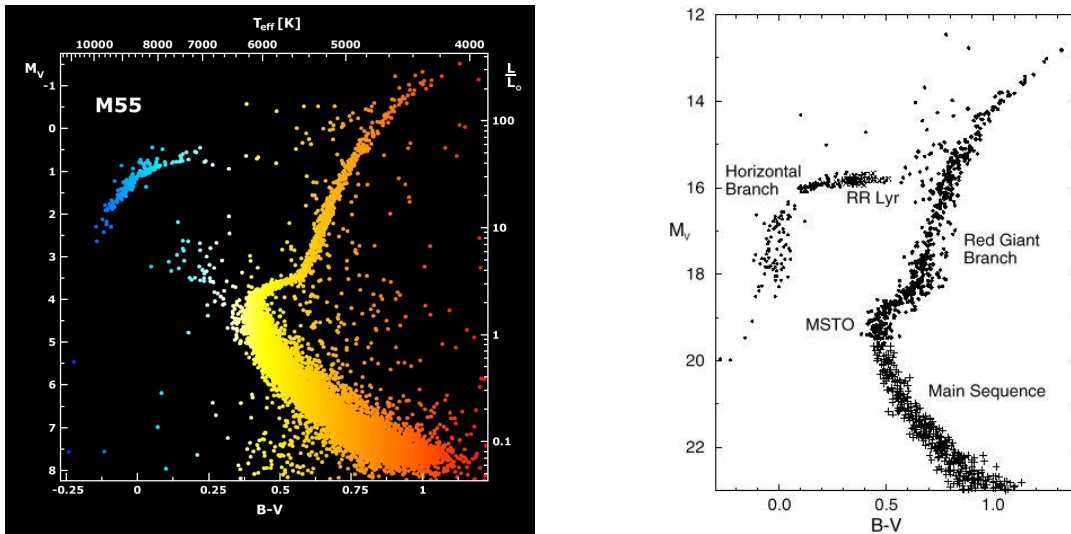
In the second part of the project, DOLPHOT was used to run photometry on ground-based data from the Gemini North telescope. This is an experimental work as DOLPHOT has never been used to carry out photometry on ground-based data. The experimental work is conducted to see if DOLPHOT can successfully run the ground-base data and give a decent CMD. Clusters used for this experiment are H25, HEC6, HEC13, PA16 and SDSS-G. These clusters can be found in the stellar halo or the outskirts of M31.



M31: The Andromeda Galaxy with credits to Jacob Bers (Bersonic Astronomy)

# Colour-Magnitude Diagrams

The Colour-Magnitude diagram (CMD) is a variation of the Hertzsprung-Russell Diagram (HR-Diagram), specially used to study star clusters. A CMD is produced by plotting a graph of the magnitudes of stars against their colours. There are two types of magnitude: absolute and apparent. The apparent magnitude tells us how bright the star is as seen from earth while the absolute magnitude tells us the brightness of the star if it were at a standard distance of 10 parsecs (32.6 light-years). Hence the absolute magnitude gives information on the distance whereas the apparent magnitude fails to do so. As for the colour, it is represented by the difference between the magnitudes of 2 different filters. The apparent magnitude can simply be converted into absolute magnitude by addition of a constant (method of conversion shown later in the "Stellar Isochrones - Distance Calculation Method" section). As a result, the calculated colour is the the same regardless of which type of magnitude used as the constant gets cancelled off when calculating the difference in the magnitudes. Now, let us take  $M_B$  as the absolute magnitude of a star and the blue-filter and likewise,  $M_V$  in the visible-filter. So since the magnitude works on an inverse scale, a negative value of colour =  $M_B - M_V$  (shown as B-V on Figure 1) implies that the star appears to be more blue than red. To give a clearer picture of how a CMDs works, the CMD of globular cluster M55 adapted from NASA is shown in Figure 1a.



(a) CMD with temperature and luminosity axes. (b) CMD with stages of stellar evolution labelled.

Figure 1: Crowding vs. Apparent Magnitude Plot

The CMD is a useful tool for classifying stars in a cluster. Most of the stars in a relatively young cluster will be found in the Main Sequence (see Figure 1b). Stars burn off hydrogen fuel to overcome strong gravitational forces within themselves, hence as stars age, the hydrogen fuels within them are being used up. As the hydrogen fuel start running low, stars will turn off the Main Sequence at the Main Sequence Turn Off point (MSTO) into the red giant branch (RGB). Typically, massive stars will burn off hydrogen fuel more quickly compared to smaller ones and hence approach the RGB more quickly than smaller stars. This is an explanation to why the RGB may not be observable yet in the CMDs of clusters with less massive stars. The RGB also provides information on the metallicity of the cluster. The metallicity is defined to be the logarithm of the star's hydrogen to iron ratio compared to that of the sun:

$$[Fe/H] = \log_{10} \left( \frac{N_{Fe}}{N_H} \right)_{star} - \log_{10} \left( \frac{N_{Fe}}{N_H} \right)_{sun} \quad (1)$$

A RGB slanted more towards the positive colour scale implies a more metal rich cluster and vice versa. Metal rich stars tend to appear redder because metals tend to absorb more blue wavelengths than red ones. The metallicity is a good probe of the age of a cluster. This is because astronomers believe that as stars evolve overtime, they become more metal rich. A possible explanation to this theory is that most of the initial mass of a star is returned to the interstellar medium (ISM) as it dies. Since stars sustain themselves by fusing elements together, metal rich elements accumulate overtime in the ISM as more star dies. Then as younger stars start to form again in the ISM, they will pick up these metallic elements and hence its initial state is more metal rich than the previous generation of stars, leading to more metallic star clusters being formed.

Progressing from the RGB, stars eventually evolve to reach the tip of the RGB which is where the brightest and reddest stars are found. According to the paper "Tip of the Red Giant Branch" by Michele Bellazzini <sup>1</sup>, the luminosity of the brightest stars at the tip of the RGB can be used as a standard candle. A standard candle is an object in the sky whose absolute magnitude is known. By taking the brightness of any cosmological object relative to the standard candle, the distance of the object from earth can be estimated using the inverse square law. Stars move away from the tip of the RGB into the horizontal branch (HB) as hydrogen fuels are completely used up. The stars will have reduced in size such that their cores become hot enough to allow nuclear fusion of helium and other heavier elements. Here in the HB, stars are then powered by helium rather than hydrogen fusion. Some stars in the HB are periodic variable star and they are known for their fluctuating apparent brightnesses as observed from earth. These stars are called the RR Lyrae variables which were historically named after RR Lyrae, the brightest variable star in the Lyra constellation. With known distances and brightnesses, RR Lyrae variables are typically used as standard candles and are able to measure galactic distances. Hence its function is analogous to that of the stars in the tip of the RGB.

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<sup>1</sup>"The Tip of the Red Giant Branch" by M. Bellazzini, Published 13th November 2007.

**Link:** <https://arxiv.org/abs/0711.2016>

# The Hubble Space Telescope



Figure 2: Image of the Hubble Space Telescope in Orbit. (Adapted from ESA Hubble Space Telescope [website](#))

## 1 What is Hubble Space Telescope?

The Hubble Space Telescope (HST) is a space-based telescope orbiting the Earth, supported by an international collaboration between NASA and the European Space Agency (ESA). The HST captures images of astronomical objects above the atmosphere of the Earth. Thus, light entering the telescope is free from the Earth's atmospheric distortion and images taken by the telescope are of higher resolution. These unperturbed images taken by the HST allow astronomers to make more accurate measurements of the universe and is nonetheless beneficial to scientific researches. The HST consists of a group of instruments, each of which are specially engineered to investigate the universe in their own way. The six main instruments according to [Hubble Site](#) powered by NASA are the Cosmic Origins Spectrograph (COS), Near Infrared Camera and Multi-Object Spectrometer (NICMOS), Space Telescope Imaging Spectrograph (STIS), Fine Guidance Sensor (FGS), Advanced Camera for Surveys (ACS) and Wide-Field Camera 3 (WFC3). The configuration of these instruments is shown in Figure 3 on the next page.

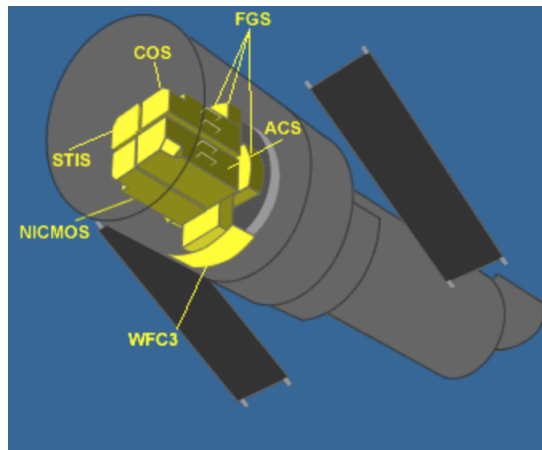


Figure 3: Science Instruments on the HST. (Adapted from NASA's Goddard Space Flight Centre [website](#))

## 2 About the Advanced Camera for Surveys (ACS)

The ACS replaced the Faint Object Camera (FOC) which has a smaller field of view, in year 2002. The ACS consists of three cameras, namely the wide field camera, the high-resolution camera and the solar blind camera. With these advanced cameras, the ACS is capable of capturing light of wavelengths ranging from the near-infrared to the ultraviolet. This broad range of wavelengths made the ACS useful for probing the distant universe, also making it an essential tool for studying galactic archaeology. A section of this project will make use of the archived images taken by the ACS to investigate the properties of the globular cluster PA02 in M31.

## 3 About Wide-Field Camera 3 (WFC3)

In 2009 during Service Mission 4, the old Wide-Field and Planetary Camera 2 (WFPC2) was removed from the HST and was replaced by WFC3. This instrument has a wider field of vision and higher resolution imaging compared to the WFPC2. It also has 63 UVIS channel filters and 17 IR channel filters<sup>2</sup> which contributes to the camera's ability to see things across a broader range of wavelengths from near infrared light to near ultraviolet radiation<sup>3</sup>.

The WFC3 has been used to explore star clusters and galaxies located at the remote ends of the universe. One of the well-known globular star clusters of Andromeda - Mayall II (or G1) was imaged by the WFC3. This project will involve detailed investigation of this cluster.

<sup>2</sup>Hubble Space Telescope WFC3 filters, STScI (Space Telescope Science Institute).

**Link:** [http://www.stsci.edu/hst/wfc3/ins\\_performance/filters/](http://www.stsci.edu/hst/wfc3/ins_performance/filters/)

<sup>3</sup>Wide-Field Camera 3: Seeing more - Seeing deeper. Hubblesite managed by NASA.

**Link:** [http://hubblesite.org/the-telescope/nuts\\_and\\_bolts/instruments/wfc3/](http://hubblesite.org/the-telescope/nuts_and_bolts/instruments/wfc3/)

# DOLPHOT - The ACS Module

## 1 DOLPHOT - Stellar Photometry

DOLPHOT is Stellar Photometry packaged designed by Andrew Dolphin to analyse data collected by the HST. According to Dolphin's publication<sup>4</sup>, DOLPHOT is a variation of HSTphot which was designed specifically to analyse data collected by the Wide Field and Planetary Camera 2 (WFPC2) on the HST. Both HSTphot and DOLPHOT works in the same way such that they are Point-Spread Functions (PSF) fitting packages which will give an output that consists of information on each PSF found. Output information include the PSF's position, instrumental magnitude, crowding, sharpness and many more. These two packages differ in the sense that HSTphot was designed specifically for the WFPC2 while DOLPHOT has a few modes, each of which are specially calibrated for specific cameras. The PSF-fitting packages are available for the ACS and WFC3 modules, as well as for the older cameras on the HST: WFPC2 and WFIRST. For each of these modules, there are a variety of PSF modules available. The ACS module includes a range of PSF wavelengths for the ACS/WFC, ACS/HRC and ACS/WFC with Anderson Core fittings. Similarly, a range of PSF wavelengths is also available for the WFC3/UVIS, WFC3/IR, their respective Anderson Core fittings, as well as for the WFPC2 and WFIRST. More information can be found through DOLPHOT's official [website](#).

To be conversant with the procedures of running DOLPHOT on ACS data, a test run is done on the globular cluster PA02 (PAndAS-02) located at the outskirts of M31 (approximately projected at a distance of 115kpc). The ".fits" file data of this cluster is obtained through the HST public archive<sup>5</sup>. Images of this cluster were taken using 2 different filters: the Visible (F606W PSF) filter and the Infrared (F814W PSF) filter. The data set retrieved from the HST archive consists of the FLC and DRC data types. FLC data are calibrated, flat-fielded and Charge-Transfer Efficiency (CTE) corrected; while the DRC data is corrected using pixel-based CTE according to STScI<sup>6</sup>. The DRC data is commonly known to be "drizzled", meaning that geometric distortion has been removed from the flat-field reduced data. The DRC data of the V-filter will be used as the reference frame in this test run.

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<sup>4</sup>DOLPHOT Stellar Photometry by Andrew Dolphin. Publication by Astrophysics Source Code Library, record ascl:1608.013. **Link:** <http://adsabs.harvard.edu/abs/2016ascl.soft08013D>

<sup>5</sup>Mukulski Archive for Space Telescopes: Hubble Space Telescope.

**Link:** <http://archive.stsci.edu/hst/>

<sup>6</sup>Space Telescope Science Institute: Hubble Space Telescope CTE Photometric Correction Formula.

**Link:** [http://www.stsci.edu/hst/acs/analysis/cte\\_formula/](http://www.stsci.edu/hst/acs/analysis/cte_formula/)

## 2 DOLPHOT Procedure for ACS data

(Note: A set of python code was generated to run this procedure)

1. Prepare **text file containing parameters** suitable for DOLPHOT/ACS (see Figure 4 on the next page). Parameters were adapted from [DOLPHOT 2.0 Manual](#) and [DOLPHOT 2.0 ACS Manual](#).
2. Ensure that the **parameter text file** is in the **directory** containing the targeted dataset
3. **Run the "acsmask"** on all the ACS images in ".fits" format
4. If data are **drizzled ACS/WFC** images, **use "splitgroups"** to separate images into two chips with the format: .chip1.fits and .chip2.fits.
5. **Run "calcsky"** on all .fits file with names containing \*.chip1.\* or \*.chip2.\*.
6. **Run DOLPHOT**

### Suggested *Terminal* command lines:

- **Run "acsmask":**  
~/path/to/dolphot2.0/bin/acsmask \*.fits
- **Use "splitgroup":**  
~/path/to/dolphot2.0/bin/splitgroups \*.fits
- **Run "calcsky":**  
~/path/to/dolphot2.0/bin/calcsky (filename).chip1 15 35 -128 2.25 2.00  
\*Repeat for all files of .chip 1 and .chip2  
*\*NB: Numbers after file name are "calcsky" parameters*
- **Run DOLPHOT:**  
~/path/to/dolphot2.0/bin/dolphot PA02.phot -pACS.param  
*\*NB: PA02.phot is the output text file*  
*\*NB: ACS.param is the parameter text file mentioned in step 1 and 2*



```
photsec =
Nimg = 12
img0_file = jbpk13010_drc.chip1
img1_file = jbpk13aqq_flg.chip1
img2_file = jbpk13aqq_flg.chip2
img3_file = jbpk13bfq_flg.chip1
img4_file = jbpk13bfq_flg.chip2
img5_file = jbpk13bnq_flg.chip1
img6_file = jbpk13bnq_flg.chip2
img7_file = jbpk13bsq_flg.chip1
img8_file = jbpk13bsq_flg.chip2
img9_file = jbpk13bwq_flg.chip1
img10_file = jbpk13bwq_flg.chip2
img11_file = jbpk13c0q_flg.chip1
img12_file = jbpk13c0q_flg.chip2
#
img_shift = 0 0
img_xform = 1 0 0
img_aprad = 10
img_apsky = 15 25
img_RAper = 3
img_RChi = 2.0
img_RSky = 15 35
img_RSky2 = 4 10
img_RPSF = 10
UseWCS = 1
ACSuseCTE = 1
ACSpsfType = 0
InterpPSFlib = 1
FlagMask = 4
PSFPhot = 1
PSFPhotIT = 3
FitSky = 2
SkipSky = 2
SkySig = 2.25
SecondPass = 5
SearchMode = 1
SigFind = 5.0
SigFindMult = 0.85
SigFinal = 5.0
MaxIT = 25
CombineChi = 0
NoiseMult = 0.10
FSat = 0.999
ApCor = 1
Force1 = 0
Align = 2
AlignIter = 3
Rotate = 1
RCentroid = 2
PosStep = 0.25
dPosMax = 2.5
RCombine = 1.5
SigPSF = 5.0
PSFres = 1
psfoff = 0.0
VerboseData = 1
```

Figure 4: **Suggested parameter text file for the ACS Module of DOLPHOT.** The list of image files before the parameter list tells dolphot specifically which data set to analyse and `img0_file` is the reference stacked image of the full field of view. Number of image files can be changed by changing the value `Nimg`.



# Useful Data from DOLPHOT Outputs

## 1 Filtering DOLPHOT Output

The output text file generated by running DOLPHOT is rather complex as it contains information on more than a hundred parameters of all the objects it has found. To simplify things, we will only be focusing on a handful of these parameters which have significant effect on the quality of the CMD to be plotted later.

Not all the objects found by DOLPHOT are stars or some detected stars may not good enough to give useful information, so it may be trivial to plot them into a CMD. Therefore, another set of python code is generated to filter out objects which are not suitable for plotting a CMD. Thus the final CMD should be constructed using objects which passed the conditions of the selection filter. Filter conditions include:

- **Radial distance from estimated centre point of cluster**

**Formula:**  $r = \sqrt{(x - x_0)^2 + (y - y_0)^2}$

$(x, y)$  = position of object

$(x_0, y_0)$  = position of the estimated centre of cluster

- **Object Type in all filters used**

Object type tells us if the detected object is a good star. (e.g. Object Type 1: good star, Object Type 2: star too faint for PSF determination, etc.) See [DOLPHOT 2.0 Manual](#) for more information on this.

- **Error Flag**

This parameter tells us how well the star was recovered from the given image. Error flag of zero implies excellent recovery. If error flag is higher than 4, the recovered star is not usable for data analysis and therefore it should be removed before plotting the CMD. (See [DOLPHOT 2.0 Manual](#) for more information)

- **Colour Uncertainty/Error**

Limiting the colour uncertainty may reduce the spreading of stars at faint magnitudes.

**Formula:**  $C_{Err} = \sqrt{(\delta M_V)^2 + (\delta M_I)^2}$

$\delta M_V$  = Instrumental Magnitude uncertainty for V-filter.

$\delta M_I$  = Instrumental Magnitude uncertainty for I-filter.

- **Instrumental Magnitude**

Depending on the distance of the cluster, objects over a certain magnitude are very faint and are usually not a star. Hence we want to exclude these trivial objects from the CMD. In the case of PA02, the limit is 30 for the average instrumental magnitudes of each filter. The value is obtained by averaging the magnitude measured from each of the three F606W (V) filter images, as well as the three F814W (I) filter images.

- **Crowding**

Crowding, measured in magnitudes, gives information on how much brighter a star would be if nearby stars affecting the brightness measurement are completely removed. A high crowding indicates a poorly measured brightness. Crowding occurs when the light of neighbouring objects/stars overlaps the targeted star. These objects are usually those which are present in the sky annuli used for photometry<sup>7</sup>. In stellar photometry of globular clusters, crowding is a huge problem as the stars are closely packed where light of nearby stars overlap one another. To plot a good CMD, crowding has to be minimized. Therefore, a filter has to be applied to filter out objects with high crowding.

- **Sharpness**

According to STScI<sup>8</sup>, the sharpness is a representation of the reciprocal of the number of pixels of a PSF, or in simpler words, how well a star fits into the PSF model. The value zero gives the maximum sharpness meaning that the star fits perfectly onto the PSF model. It can be used to determine the number of pixels for optimal photometric extraction. Therefore, good PSFs highly depends on their sharpness.

## 2 Plotting the CMD of PA02

Previously, we saw that the CMD is constructed by plotting the absolute magnitude against the colour. Here, the CMD for PA02 is constructed by plotting the apparent (instrumental) magnitude against colour instead, where  $\text{colour} = M_V - M_I$ . The CMD is constructed using only DOLPHOT outputs which passed the filter conditions. The list of passing conditions is given on the next page.

By observing the PA02 CMD shown in Figure 5, we are able to see the RGB after the turn-off region as well as a HB. There are two reasons why we do not see the main sequence stars in this cluster. One is that the cluster may be very old and all the stars have evolved from the main sequence, or the other reason could be that the main sequence stars are too faint to be detected by the telescope since the cluster is very far away. According to Dr. Dougal Mackey, this cluster

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<sup>7</sup>Crowded Field Photometry by New Mexico State University (NMSU) Astronomy.

Link: <http://astronomy.nmsu.edu/holtz/a535.fall08/ay535notes/node70.html>

<sup>8</sup>UVIS Optical Performance, STScI (Space Telescope Science Institute).

Link: [http://www.stsci.edu/hst/wfc3/documents/handbooks/currentIHB/c06\\_uvis07.html](http://www.stsci.edu/hst/wfc3/documents/handbooks/currentIHB/c06_uvis07.html)

has a very blue (negative colour) HB morphology. Therefore, the HB of PA02 shown in Figure 5 is not exactly complete because the V and I filters are unable to capture light in the bluer end of the electromagnetic spectrum.

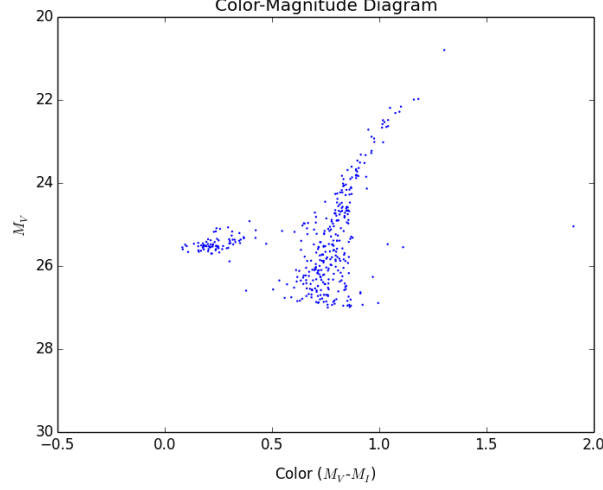


Figure 5: Colour-Magnitude Diagram of globular cluster PA02

#### Object passing conditions (i.e. Filter list):

- **Radial distance:**  $r < 400$  pixels
- **Object Type:** Type 1 only
- **Colour Uncertainty:**  $C_{Err} < 0.1$

\*Conditions below apply to all X-filters (e.g. B, V, I, R and etc.)

- **Error Flag in X-filter:**  $F_{Err_X} \leq 4$
- **Instrumental Magnitude in X-filter:**  $M_X < 30$
- **Sharpness in X-filter:**  $-0.1 < X_{sharpness} < 0.1$
- **Crowding in X-filter:**  $X_{crowding} < 0.1$

The figure below shows the (row, column) of the output data before and after the filter code is applied. The number of rows represent the number of object and the columns are simply the parameters for each object. So in this case, each object has 193 parameters.

|                                                                            |
|----------------------------------------------------------------------------|
| Initial Object Count:<br>(60079, 193)<br>Final Object Count:<br>(386, 193) |
|----------------------------------------------------------------------------|

Figure 6: Object count before and after filter conditions

Filtering the output data is very important. Otherwise the CMD produced will be a mess and will contain a lot of trivial data points. Figure 7 on the next page shows an example of CMD of PA02 plotted without filtering the dolphot output.

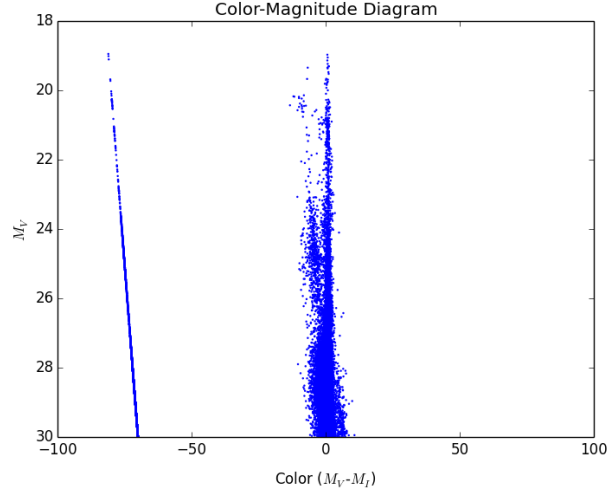


Figure 7: Colour-Magnitude Diagram constructed using unfiltered DOLPHOT output. This CMD consists of 60079 data points

### 3 Sharpness and Crowding Plots of PA02

On top of the CMD, we can also plot the PSF sharpness against its magnitude in a given filter and similarly for the crowding. The sample plots for PA02 are shown in Figures 8 and 9.

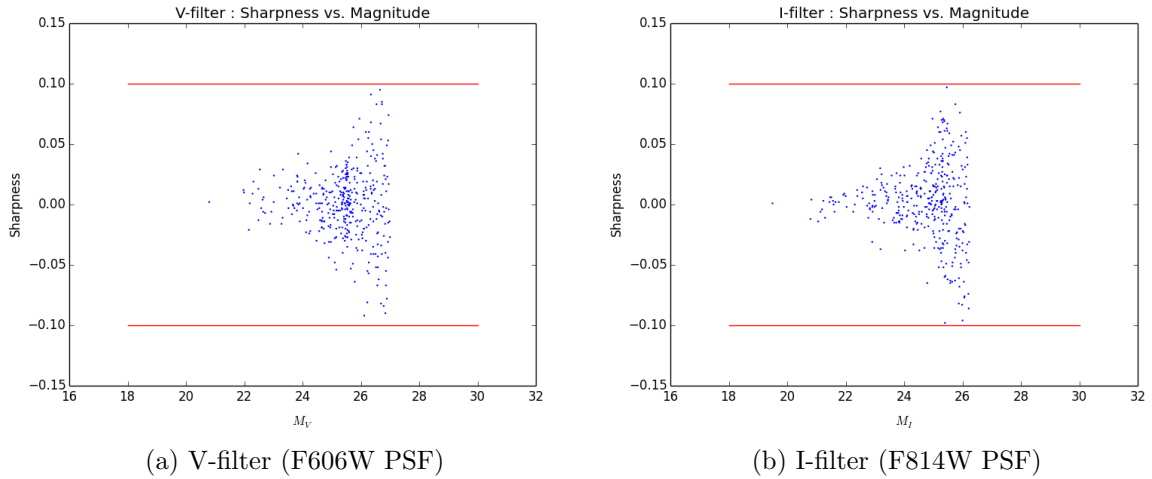


Figure 8: Sharpness vs. Apparent Magnitude Plot

#### Sharpness Plot Analysis:

Sharpness decreases for larger apparent magnitudes as expected, since larger apparent magnitudes imply fainter. Fainter stars have less well-defined PSFs and hence does not fit as well into the PSF model.

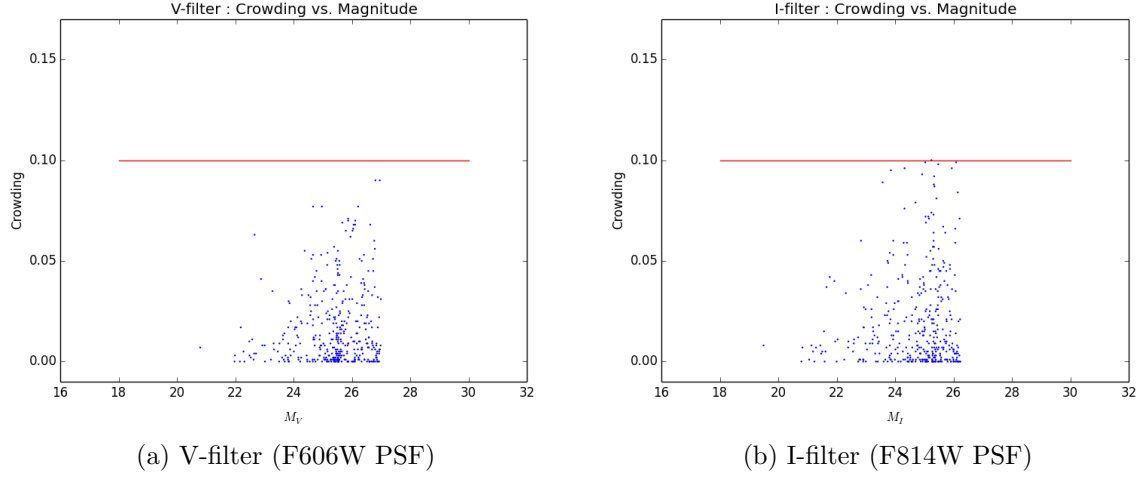


Figure 9: Crowding vs. Apparent Magnitude Plot

### **Crowding Plot Analysis:**

PA02 is a loosely packed globular cluster where crowding is relatively small and hence its photometry gave relatively good results. However at greater magnitudes, stars are fainter and logically fainter objects would be more vulnerable to any surrounding brightness. Therefore, crowding increases with magnitude as seen in the two crowding plots above.

## **4 Radial Plot of PA02**

The radial plot tells us how the population of stars vary as we move away from the approximate centre point of the cluster.

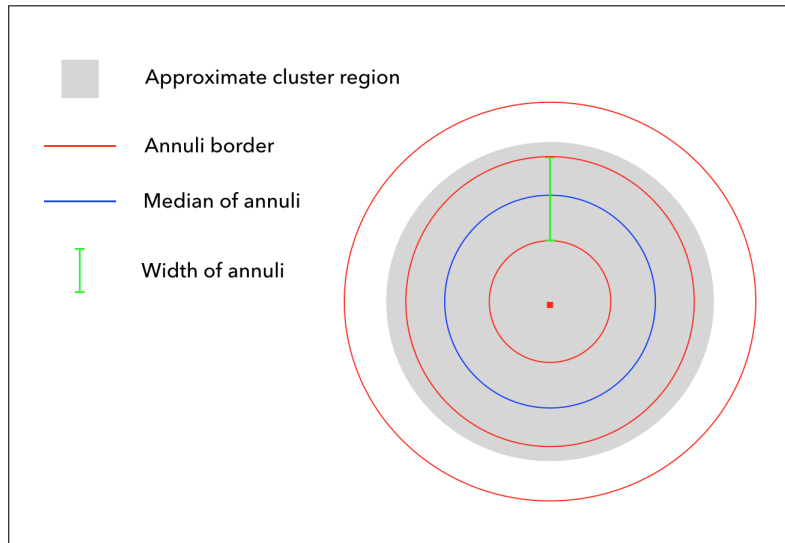


Figure 10: Method for generating a radial plot. The point right in the middle of the cluster region indicates the approximate centre of the cluster.

Figure 10 illustrated the method at which radial plot was generated. The point right in the middle indicates the approximate centre of the cluster. As an example, let's say we are trying to find the approximate number of stars at a radius of 300 pixels from the centre. An algorithm programmed using python will take two circles of radii 200 and 400 pixels to form an annuli of a size of 200 pixels about the centre, and then count the number of objects within this annuli. This number is then used as the approximate stellar population at the median of the two chosen radii, which in this case is 300 pixels as we desired. The program iterates over a range of radii given and plots a graph in the linear and logarithmic scale with number of stars in the y-axis and median radii in the x-axis.

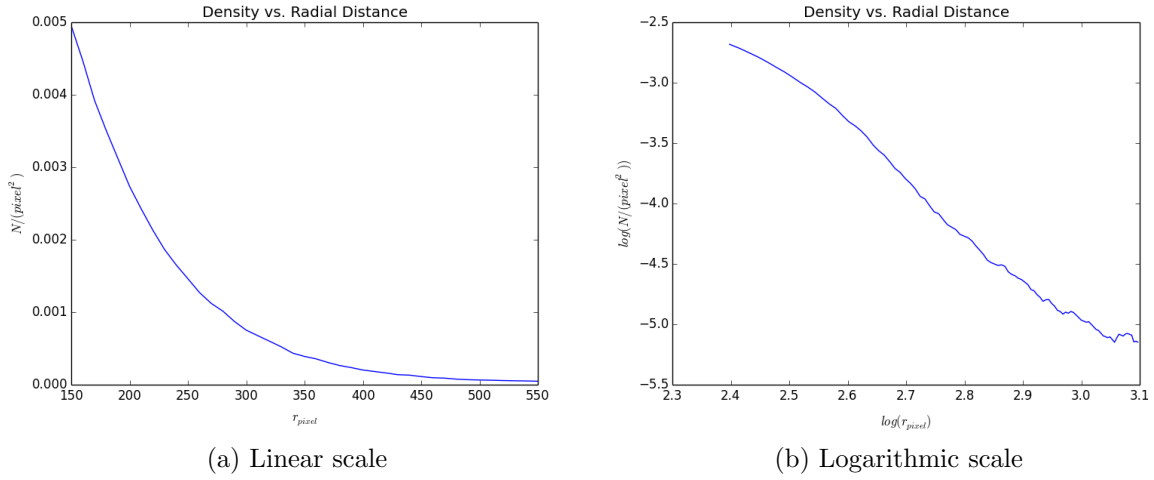
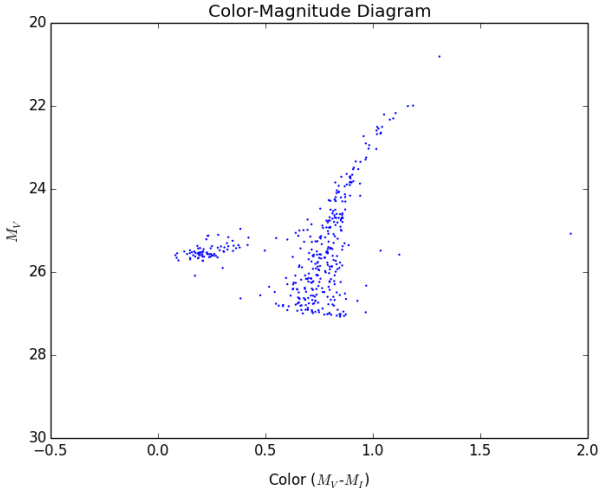
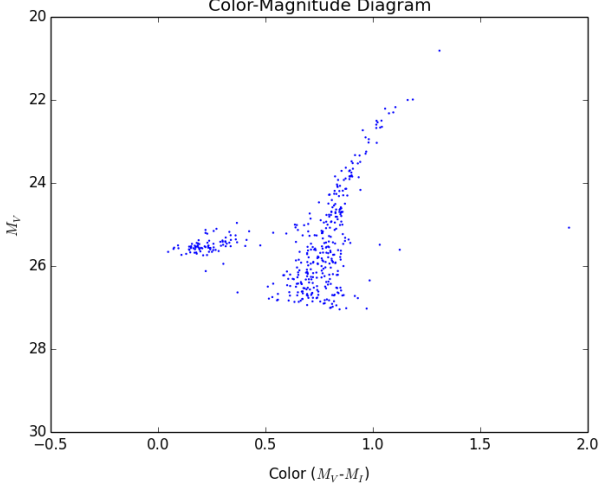


Figure 11: Radial plots of PA02 with 500 pixel annuli.

As we would expect to see, the population of stars decreases as we move away from the centre of the cluster. The slight variations at the larger radii in the logarithmic scale plot (Figure 11b) could be due to the irregularity of stellar populations at the edge of the cluster.

# DOLPHOT with Different Parameters for PA02

## 1 Colour-Magnitude Diagrams

| Trial No. | Parameters Used and Object Count                                                                                                                                                                                                                                                 | Colour-Magnitude Diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | <p>Refer to Figure 12</p> <div data-bbox="381 1133 604 1227" style="border: 1px solid black; padding: 5px;"> Initial Object Count:<br/>(53169, 193)<br/>Final Object Count:<br/>(390, 193) </div>                                                                                |  <p>A scatter plot titled 'Color-Magnitude Diagram' showing the relationship between absolute magnitude <math>M_V</math> (y-axis, ranging from 20 to 30) and color <math>(M_V - M_I)</math> (x-axis, ranging from -0.5 to 2.0). The plot displays a dense cluster of blue points, with a distinct branch curving upwards and to the right, characteristic of a stellar population's color-magnitude diagram.</p>  |
| 2         | <p>Refer to Figure 12 with <math>\text{FitSky} = 3</math> instead of <math>\text{FitSky} = 1</math>.</p> <div data-bbox="381 1711 604 1805" style="border: 1px solid black; padding: 5px;"> Initial Object Count:<br/>(72771, 193)<br/>Final Object Count:<br/>(415, 193) </div> |  <p>A scatter plot titled 'Color-Magnitude Diagram' showing the relationship between absolute magnitude <math>M_V</math> (y-axis, ranging from 20 to 30) and color <math>(M_V - M_I)</math> (x-axis, ranging from -0.5 to 2.0). The plot displays a dense cluster of blue points, with a distinct branch curving upwards and to the right, characteristic of a stellar population's color-magnitude diagram.</p> |

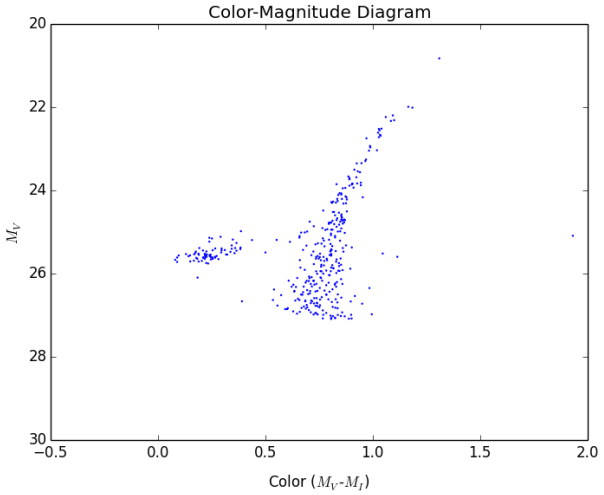
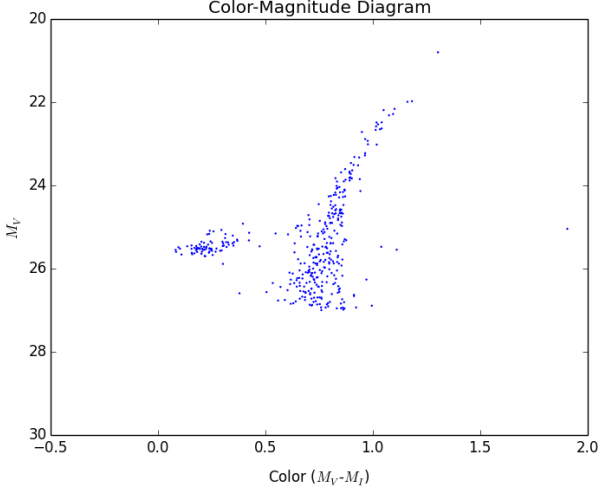
| Trial No. | Parameters Used and Object Count                                                                                                                                                                                                                  | Colour-Magnitude Diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3         | <p>Refer to Figure 12 with ACSpsfType = 0 instead of ACSpsfType = 1 which uses the PSF library with Anderson PSF core.</p> <div data-bbox="379 824 608 920"> Initial Object Count:<br/>(54407, 193)<br/>Final Object Count:<br/>(374, 193) </div> |  <p>A scatter plot titled 'Color-Magnitude Diagram' showing the relationship between absolute magnitude <math>M_V</math> (y-axis, ranging from 20 to 30) and color <math>(M_V - M_I)</math> (x-axis, ranging from -0.5 to 2.0). The plot displays a dense cluster of blue points representing stars, with a clear main sequence and a branch extending towards higher magnitudes and colors.</p>   |
| 4         | <p>Refer to Figure 4</p> <div data-bbox="379 1249 608 1346"> Initial Object Count:<br/>(60079, 193)<br/>Final Object Count:<br/>(386, 193) </div>                                                                                                 |  <p>A scatter plot titled 'Color-Magnitude Diagram' showing the relationship between absolute magnitude <math>M_V</math> (y-axis, ranging from 20 to 30) and color <math>(M_V - M_I)</math> (x-axis, ranging from -0.5 to 2.0). The plot displays a dense cluster of blue points representing stars, with a clear main sequence and a branch extending towards higher magnitudes and colors.</p> |

Table 1: Colour-Magnitude Diagrams of PA02 generated using the outputs of several DOLPHOT runs. Each DOLPHOT run was conducted using different parameters.

#### General Observations from Table 1:

- Trial 2 using FitSky = 3 has the highest initial and final object count.
- But generally, all the CMDs produced using DOLPHOT ACS outputs of different parameters are very similar and describes the same morphology for PA02.



```

#image-specific parameters
photsec =
Nimg = 12
img0_file = jbpk13010_drc.chip1
img1_file = jbpk13aqq_flg.chip1
img2_file = jbpk13aqq_flg.chip2
img3_file = jbpk13bfq_flg.chip1
img4_file = jbpk13bfq_flg.chip2
img5_file = jbpk13bnq_flg.chip1
img6_file = jbpk13bnq_flg.chip2
img7_file = jbpk13bsq_flg.chip1
img8_file = jbpk13bsq_flg.chip2
img9_file = jbpk13bwq_flg.chip1
img10_file = jbpk13bwq_flg.chip2
img11_file = jbpk13c0q_flg.chip1
img12_file = jbpk13c0q_flg.chip2
#
#photometry-specific parameters
img_shift = 0 0
img_xform = 1 0 0
img_aprad = 10
img_apsky = 15 25
img_RAper = 4
img_RChi = 2.0
img_RSky = 15 35
img_RPSF = 10
UseWCS = 1
ACSuseCTE = 0
ACSpsfType = 0
InterpPSFlib = 1
FlagMask = 4
PSFPhot = 1
PSFPhotIT = 3
FitSky = 1
SkipSky = 1
SkySig = 2.25
SecondPass = 5
SearchMode = 1
SigFind = 5.0
SigFindMult = 0.85
SigFinal = 5.0
MaxIT = 25
CombineChi = 0
NoiseMult = 0.10
FSat = 0.999
ApCor = 1
Force1 = 0
Align = 2
AlignIter = 3
Rotate = 1
RCentroid = 2
PosStep = 0.25
dPosMax = 2.5
RCombine = 1.5
SigPSF = 5.0
PSFres = 1
psfoff = 0.0
VerboseData = 1

```

Figure 12: DOLPHOT 2.0 ACS Module modified parameter example.

## 2 Sharpness and Crowding

| Trial No. | Sharpness and Crowding Plots                                                                                                                                                                                                                                                                                                                        |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | <div data-bbox="502 645 917 952"> <p>V-filter : Sharpness vs. Magnitude</p> </div> <div data-bbox="933 645 1300 952"> <p>V-filter : Crowding vs. Magnitude</p> </div> <div data-bbox="502 952 917 1265"> <p>I-filter : Sharpness vs. Magnitude</p> </div> <div data-bbox="933 952 1300 1265"> <p>I-filter : Crowding vs. Magnitude</p> </div>       |
| 2         | <div data-bbox="502 1317 917 1624"> <p>V-filter : Sharpness vs. Magnitude</p> </div> <div data-bbox="933 1317 1300 1624"> <p>V-filter : Crowding vs. Magnitude</p> </div> <div data-bbox="502 1624 917 1937"> <p>I-filter : Sharpness vs. Magnitude</p> </div> <div data-bbox="933 1624 1300 1937"> <p>I-filter : Crowding vs. Magnitude</p> </div> |

| Trial No. | Sharpness and Crowding Plots                                                                                                                                                                                                                                                                                                                                   |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3         | <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;"> <p>V-filter : Sharpness vs. Magnitude</p> </div> <div style="width: 50%;"> <p>V-filter : Crowding vs. Magnitude</p> </div> <div style="width: 50%;"> <p>I-filter : Sharpness vs. Magnitude</p> </div> <div style="width: 50%;"> <p>I-filter : Crowding vs. Magnitude</p> </div> </div> |
| 4         | <div style="display: flex; flex-wrap: wrap;"> <div style="width: 50%;"> <p>V-filter : Sharpness vs. Magnitude</p> </div> <div style="width: 50%;"> <p>V-filter : Crowding vs. Magnitude</p> </div> <div style="width: 50%;"> <p>I-filter : Sharpness vs. Magnitude</p> </div> <div style="width: 50%;"> <p>I-filter : Crowding vs. Magnitude</p> </div> </div> |

Table 2: Sharpness and crowding plotted against absolute magnitude. These plots were also generated using DOLPHOT ACS outputs of different parameter runs.

**General Observations according to Table 2:**

- PSFs in Trial 1, 3, and 4 is less sharp than PSFs in Trial 2.
- PSFs are least sharp in Trial 4.
- A general trend of high crowding in more positive magnitudes for Trials 1, 3, and 4. However in Trial 2, the trend is not as obvious.
- Overall crowding is the least in Trial 4

# DOLPHOT - The WFC3 Module

## 1 Running DOLPHOT on WFC3 Data of Mayall II (G1)

As mentioned previously, DOLPHOT has PSF-fitting packages for various cameras at different wavelengths. For the PA02 data, we utilized the ACS/WFC PSF-fitting package with F814W and F606W PSF models to run DOLPHOT. As for the G1 dataset, we will be using the WFC3 package with F606W (appx. V-filter) and F438W (appx. B-filter) PSF models.

The reason for choosing G1 is because it is one of the brightest massive globular cluster located in M31. Astronomers believe that it may have some interesting properties such as having a metallicity spread across the cluster which can be observed through its CMD (see Figure 16). Also, stars within this cluster is much more compact in contrast to the loosely packed stars in PA02 (see Figure 13 and 14).

The procedures of running DOLPHOT using the WFC3 module is very similar to the procedure using the ACS module. The only differences are that we require a new parameter text file suitable for running this module and to use "wfc3mask" instead of "acsmask". The suggested parameters as per the [DOLPHOT 2.0 Manual](#) and the [DOLPHOT 2.0 WFC3 Manual](#) can be found in Figure 15.

## 2 Colour-Magnitude Diagram of G1

Once again, we use the output data of DOLPHOT to produce the CMD. The procedures are exactly the same as the ACS module, except that the colour is now  $M_B - M_V$  (or strictly speaking: Colour =  $M_{F606W} - M_{F438W}$ , but either way works since it is just a notation).

The filter conditions are also kept the same, except that the instrumental magnitude in both filters are now limited to less than 28 rather than 30. This cluster contains a higher proportion of brighter stars compared to PA02. Therefore, we want to be more strict in terms of selecting the "good stars" which will contribute to the G1 CMD in Figure 16.

There are a few things that are observable from the CMD of G1. The RGB of this cluster is very slanted (i.e. extends into the redder side of the colour scale). This implies that the cluster may be quite metal rich, although we cannot tell exactly what its metallicity is. Also, the tip of the RGB in this cluster is noticeably thicker, in other words, there is a greater colour distribution for a given magnitude.

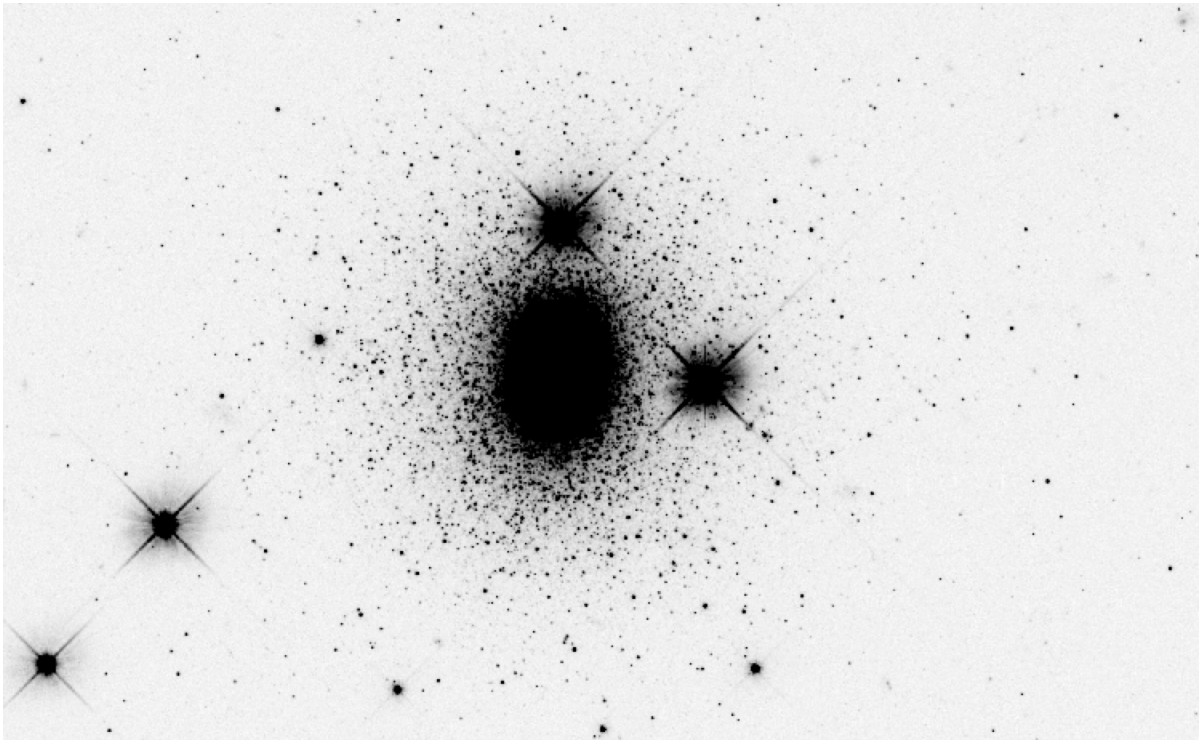


Figure 13: Image of G1 viewed using `ds9`

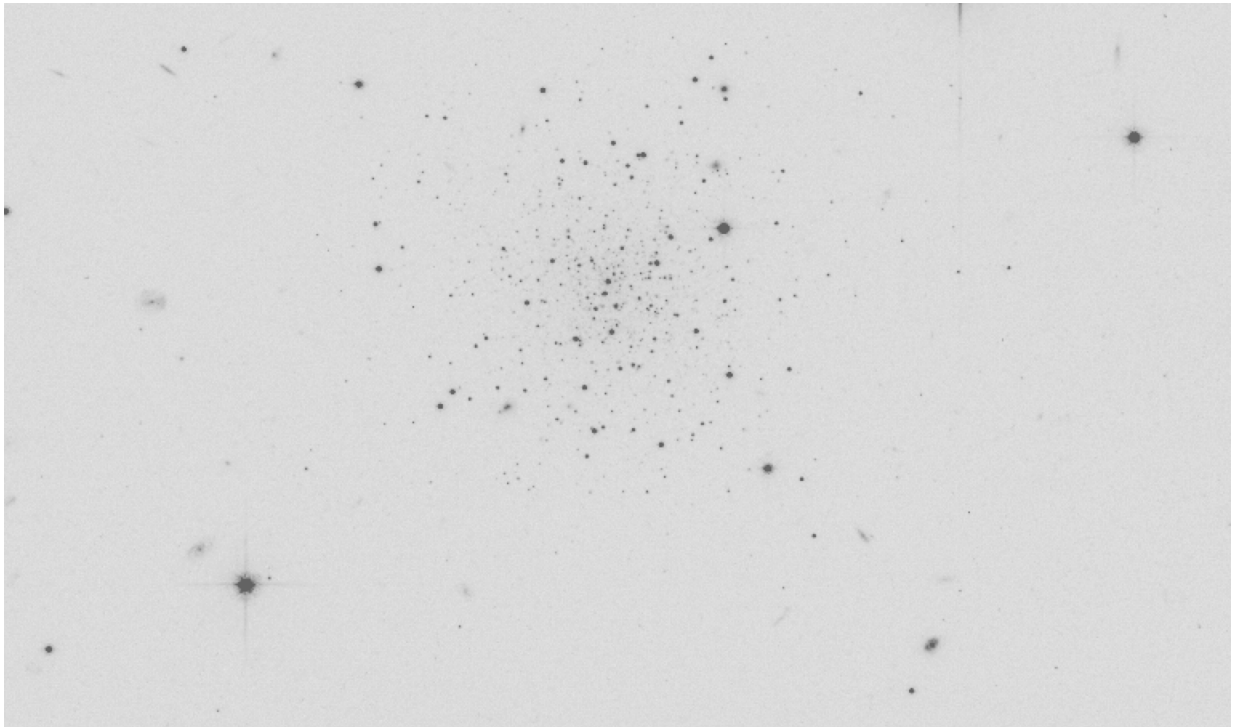


Figure 14: Image of PA02 viewed using `ds9`

```

photsec =
Nimg = 24
img0_file = ibis04020_drc.chip1
img1_file = ibis04u0q_flc.chip1
img2_file = ibis04u0q_flc.chip2
img3_file = ibis04u4q_flc.chip1
img4_file = ibis04u4q_flc.chip2
img5_file = ibis04ufq_flc.chip1
img6_file = ibis04ufq_flc.chip2
img7_file = ibis04upq_flc.chip1
img8_file = ibis04upq_flc.chip2
img9_file = ibis04tvq_flc.chip1
img10_file = ibis04tvq_flc.chip2
img11_file = ibis04txq_flc.chip1
img12_file = ibis04txq_flc.chip2
img13_file = ibis05wuq_flc.chip1
img14_file = ibis05wuq_flc.chip2
img15_file = ibis05wwq_flc.chip1
img16_file = ibis05wwq_flc.chip2
img17_file = ibis05x0q_flc.chip1
img18_file = ibis05x0q_flc.chip2
img19_file = ibis06x5q_flc.chip1
img20_file = ibis06x5q_flc.chip2
img21_file = ibis06x7q_flc.chip1
img22_file = ibis06x7q_flc.chip2
img23_file = ibis06xbq_flc.chip1
img24_file = ibis06xbq_flc.chip2
#
img_shift = 0 0
img_xform = 1 0 0
img_aprad = 0.5
img_apsky = 15 25
img_RAper = 3
img_RChi = 2.0
img_RSky = 15 35
img_RSky2 = 4 10
img_RPSF = 13
UseWCS = 1
WFC3useCTE = 1
WFC3IRpsfType = 0
WFC3UVISpsfType = 0
InterpPSFlib = 1
FlagMask = 4
PSFPhot = 1
PSFPhotIT = 3
FitSky = 2
SkipSky = 2
SkySig = 2.25
SecondPass = 5
SearchMode = 1
SigFind = 5.0
SigFindMult = 0.85
SigFinal = 5.0
MaxIT = 25
CombineChi = 0
NoiseMult = 0.10
FSat = 0.999
ApCor = 1
Force1 = 0
Align = 2
AlignIter = 3
Rotate = 1
RCentroid = 2
PosStep = 0.25
dPosMax = 2.5
RCombine = 1.5
SigPSF = 5.0
PSFres = 1
psfoff = 0.0
VerboseData = 1

```

Figure 15: **Suggested parameter text file for the WFC3 Module of DOLPHOT.** Notice that the number of images have been doubled when compared to the number of images in the ACS parameter text file. This is because we now have 12 images, each of which are split into 2 chips (single-chip format) by "*splitgroup*". Images with names "ibis05..." or "ibis06..." were taken in the F606W filter; Images with "ibis04..." were taken in the F438W filter.

According to Dr. Mackey, G1 has an intrinsic metallicity spread. This can be an indication that the cluster contains stars of different generations implying that the cluster may have formed over a large time scale. Formation of star clusters over a relatively long period of time may result in the variation of iron abundances within each individual star. This metallicity spreading characteristic is only observed in a handful of massive clusters, which makes G1 a scientifically intriguing globular cluster.

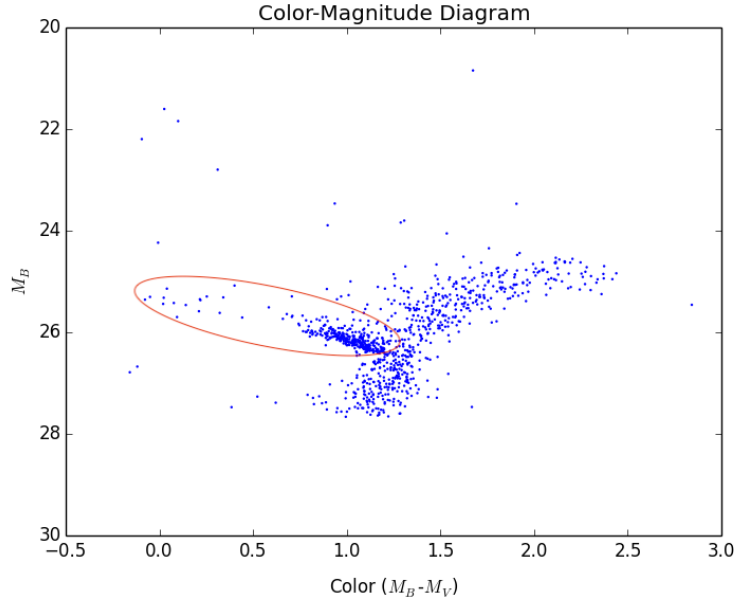


Figure 16: **Colour-Magnitude Diagram of G1.** The region enclosed within the red circle is the horizontal branch.

### 3 Sharpness and Crowding Plots of G1

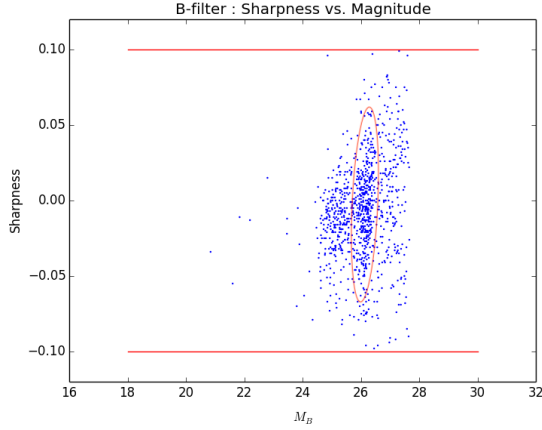
#### Sharpness Plot Analysis:

G1 is a more compact cluster and a lot of stars may overlap. As a result, there are a lot more stars that fit less well onto the PSF model hence their sharpness isn't as great as that of the stars in PA03. Also, another interesting thing to realize is that the stars of the HB can be seen on this plot - the distinct region that lies at about magnitude of 26 in the B-filter and at 25 in the V-filter (both circled in red in Figure 17). This feature was not obvious in the PA02 plots as there are lesser stars in HB of that cluster and they are more scattered.

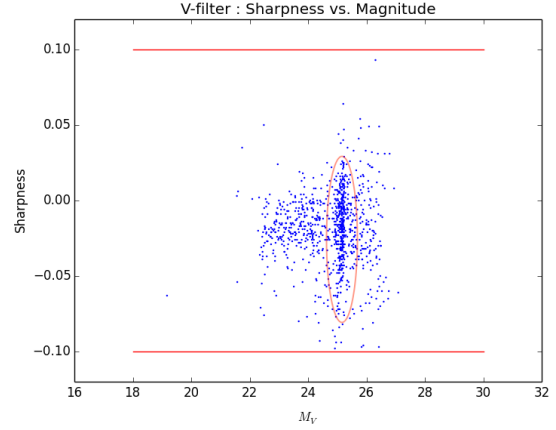
#### Sharpness Plot Analysis:

Crowding should be larger for more compact clusters as neighbouring PSFs will overlap the targeted PSF, which is exactly what we see in the G1 crowding plot. There was an obvious trend in PA02 where fainter stars of larger magnitude is more likely to be affected by crowding.



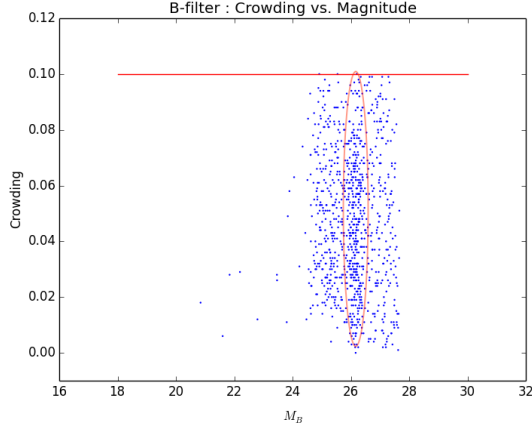


(a) B-filter (F438W PSF)

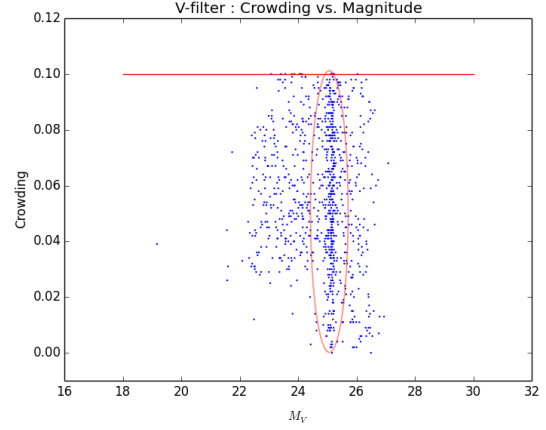


(b) V-filter (F606W PSF)

Figure 17: Sharpness vs. Apparent Magnitude Plot



(a) B-filter (F438W PSF)



(b) V-filter (F606W PSF)

Figure 18: Crowding vs. Apparent Magnitude Plot

However, in the G1 clusters, regardless of how small the magnitude is, there is still a relatively large crowding due to the overlapping of light from nearby and closely packed stars.

## 4 Radial Plots of G1

According to Figure 19 on the next page, we can see an obvious decreasing trend in stellar population as we move away from the centre of the cluster which is exactly the same trend observed in PA02 radial plots. The variation at large radius in the logarithmic radial plot of G1 has the same explanation as to that of PA02. However, the linear scale plot is not smooth at small radii unlike the PA02 linear radial plot. This sawtooth-looking region is due to the severe crowding at the centre of the region where DOLPHOT could not detect individual stars as they

are all appear to be clumped together in the image. In reality, stellar population density should be the greatest near the centre of the G1 cluster.

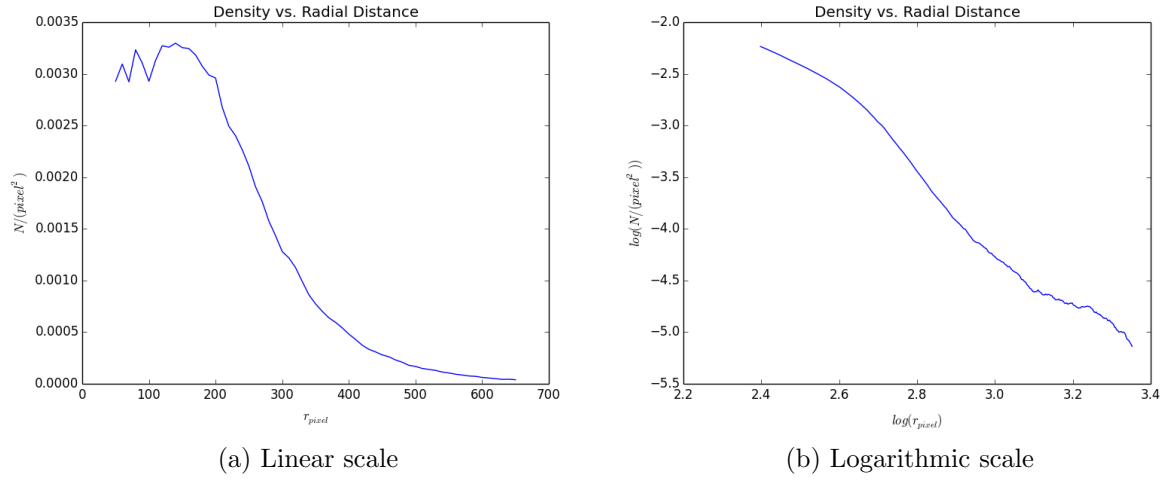


Figure 19: Radial plots of G1 with 500 pixel annuli.

## Comparing G1 and PA02

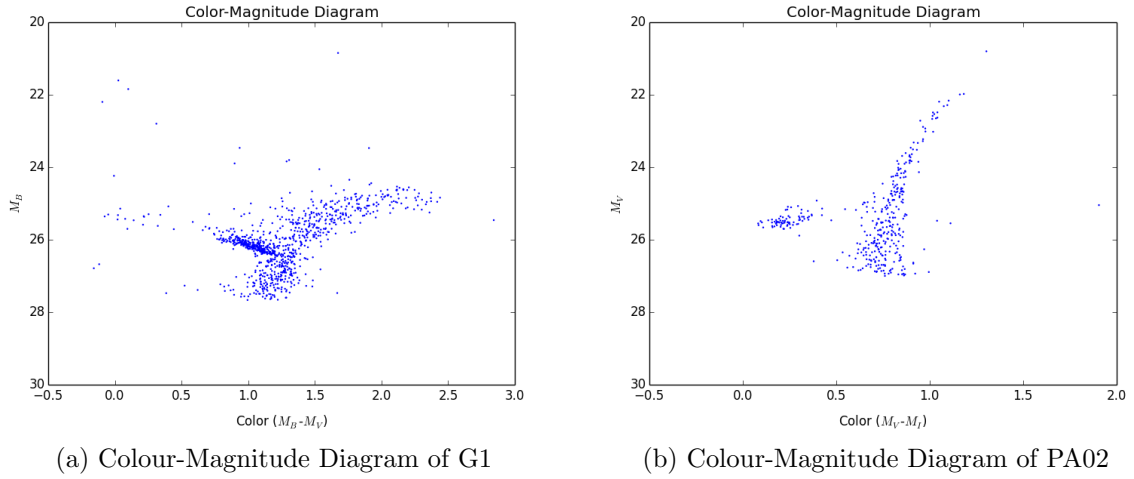


Figure 20: Colour-Magnitude Diagrams

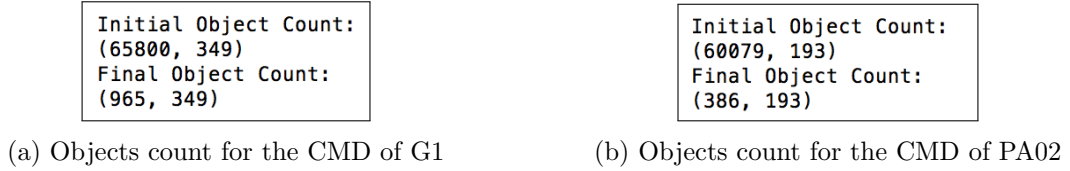


Figure 21: Colour-Magnitude Diagrams

| CMD Region             | Comments                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Red Giant Branch (RGB) | Recall that the steepness of the RGB is a proxy for metallicity. Here, we observe that the RGB of G1 is more slanted towards the right (i.e. redder on the colour scale). This implies that G1 has a higher metallicity compared to PA02. Most of the blue light have been absorbed by the metal rich elements within G1, hence stars within the cluster appear redder than those of PA02. |

| CMD Region             | Comments                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tip of the RGB (TRGB)  | Comparing the TRGB of PA02 and G1, we can observe that the tip is less well-defined in G1 due to the metallicity spread as previously mentioned. To elaborate further on the previous guesses on the metallicity spread phenomenon, the paper "Mayall II = G1 in M31" <sup>9</sup> (G. Meylan et al. (2001)) suggested that it could be due to stars within the cluster having different chemical composition. However, the actual explanation to this spread in G1 is still an ongoing research topic. |
| Horizontal Branch (HB) | The HB of PA02 has a much bluer morphology compared to that of G1. The two HBs are also slanted in a different directions. The differences between the HBs could be due to the second parameter effect which will be described in more detail later. Rich et al. (1996) also pointed that the bluer extension of G1's HB could once again be due to the wide range of chemical abundance in the stars of this cluster.                                                                                  |
| Object Count           | By comparing Figure 13 and 14, it is expected that the object count for G1 is higher than PA02. G1 is a massive globular cluster in M31, hence it is logical that the number of objects passing the filter conditions in G1 is higher than that of the loosely packed globular cluster, PA02.                                                                                                                                                                                                           |

Table 3: Comparing globular clusters PA02 and G1.

<sup>9</sup>Mayall II = G1 in M31: Giant Globular Cluster or Core of a Dwarf Elliptical Galaxy?  
Link: <https://arxiv.org/pdf/astro-ph/0105013.pdf>

# Stellar Isochrones

## 1 Introduction to Stellar Isochrones

Stellar Isochrones are plots on the HR-Diagram representing a group of stars of similar ages. They are generated using theoretical calculations of ageing stellar populations for the purpose of modelling single-star stellar evolutions over a range of stellar masses. Since isochrones are produced theoretically, it contains important information such as the cluster's metallicity, its age and in some cases the alpha-element to iron abundance (i.e.  $[\alpha/\text{Fe}]$ ). The abundances of  $\alpha$ -elements: oxygen (O), neon (Ne), magnesium (Mg), silicon (Si), calcium (Ca) and Titanium (Ti), relative to Iron (Fe) are essential for determining the age of the cluster as these two parameters are interrelated<sup>10</sup>.

Generally, stellar isochrones are observational models of stellar population<sup>11</sup>. In simpler words, they are simulated CMDs of a star cluster with a known set of parameters, which one would expect to see. However, there is a significant difference between CMDs and stellar isochrones. In CMDs, absolute magnitudes are plotted against colour; in stellar isochrones, apparent magnitude replaces the absolute magnitude. An example of stellar isochrone is shown in Figure 22.

Stellar isochrones are widely used to determine the physical properties of unfamiliar clusters. The method for doing so is called isochrone fitting. This technique will be applied to both the globular clusters: PA02 and G1, which have undergone photometry, to determine the distance from earth to these clusters.

## 2 Isochrone Fitting

Isochrone fitting is a commonly used method to constrain the possible physical properties that a stellar population could have, such as its age, distance, metallicity and etc. There are still ongoing work to formulate a mathematical method which can be used to conduct isochrone fitting but many developed work are still being argued over. David Valls-Gabaud has published a [paper](#) in 2014, on an algorithm called the Bayesian Isochrone Fitting, but many astronomers

---

<sup>10</sup>Chapter 7: Alpha-Element to Iron Overabundance by Rosaria Tantalò from the Padova GALADRIEL.

Link: [http://www.astro.unipd.it/galadriel/Tantalò/Thesis/chapter\\_7.pdf](http://www.astro.unipd.it/galadriel/Tantalò/Thesis/chapter_7.pdf)

<sup>11</sup>Observational Modelling of Stellar Populations by Justin Adkins of Eastern Kentucky University (EKU).

Link: <http://www.astro.wisc.edu/~adkins/index.html>

still prefer to perform isochrone fitting "by-eye". A few scientific papers like Gabriel Perren's 2015 [paper](#) and Aaron Dotter's 2011 [paper](#) were published using the traditional fitting by-eye method. However, this method depends on individual judgements. Thus ambiguity arises and the results may lead to controversies as each individual would prefer their own method of judgement. In addition to that, this method may not be suitable for fitting a large amount of data as it is tedious and time-consuming. Nevertheless, fitting by-eye still does a decent job for small datasets as some perplexed errors in formulated methods are yet to be removed.

The main isochrone generator used for the fitting of PA02 and G1 is the [MESA Isochrones and Stellar Tracks \(MIST\)](#). However, we also used isochrones generators by Léo Girardi ([Observatorio Astronomico di Padova](#)) and Aaron Dotter ([Dartmouth Stellar Evolution Database](#)) to compare the results with MIST's isochrone fitting. The figure below shows an example of a MIST isochrone.

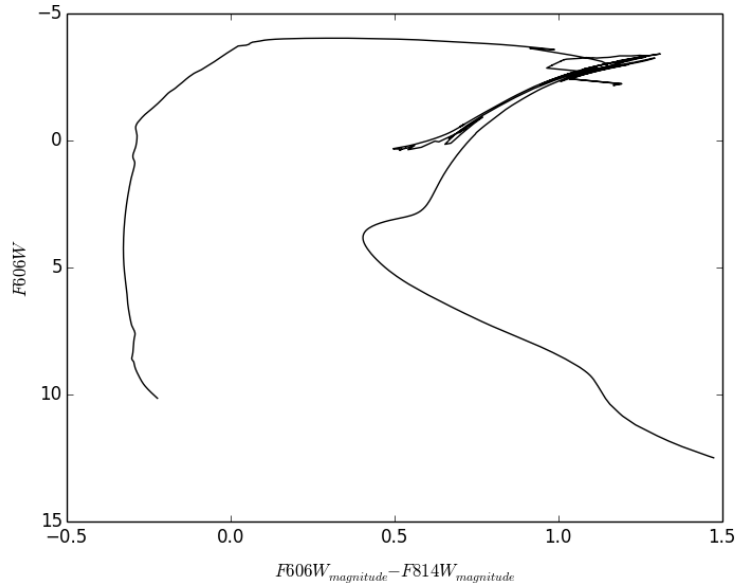


Figure 22  
 Isochrone generator: MIST  
 Synthetic Photometry: HST ACS/WFC  
 Age: 12 Gyr (12 billion years)  
 Metallicity: -1.20

The methodology for the PA02 and G1 isochrone fitting is to shift the isochrone along the x and y axes to satisfy two major conditions. Usually we would prioritize the alignment of the main sequence turn off (before the sub-giant branch), but since the turn off is not observed on the CMD of both PA02 and G1, the first condition is to ensure that the RGB of the isochrone sits in the centre of the RGB on the CMD. The second alignment condition is that the HB position of the isochrone should be approximately the same as that of the CMD. Figures [23](#) and [24](#) are a few examples of isochrone fits on the same CMD of PA02.

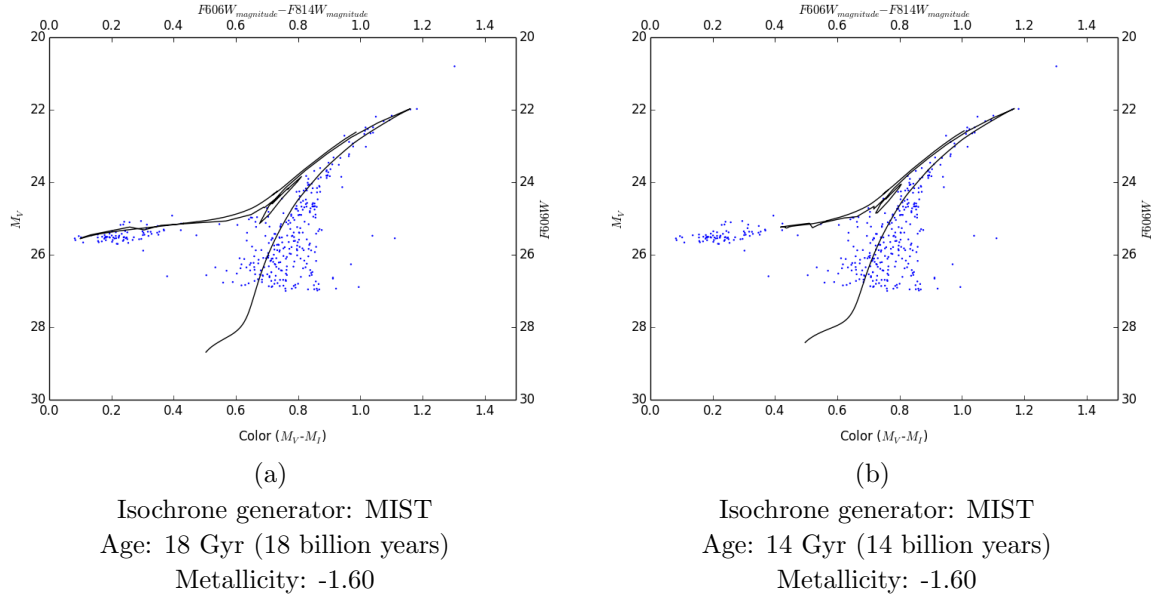


Figure 23: Examples of good isochrone fits

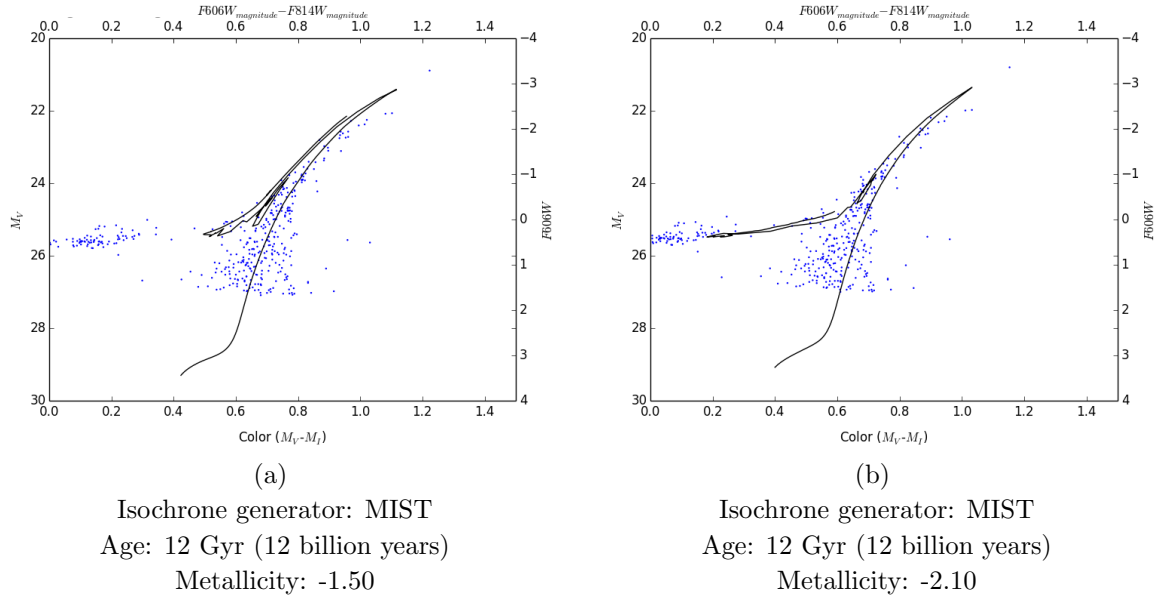


Figure 24: Examples of bad isochrone fits

All the isochrones used in Figures 23 and 24 have been filtered where all star types are removed except for those in the turn off, the RGB and the HB. The two isochrone fits in Figure 23 are examples of good isochrone fits. However, if we were to compare the two figures, Figure 23a) with an older isochrone has a better fit than that of the younger isochrone in Figure 23b) because the HB fits better on the actual CMD. Since the two isochrones are of the same metallicity but different age, it gives a hint that there is a close relationship between the age of the cluster and the morphology of the HB. This once again has something to do with the second parameter problem.

In contrast to Figure 23, Figure 24 shows two bad examples of isochrone fits. The isochrone on Figure 24a does not fit well onto the HB and the tip of the RGB, although it does start off lying in the middle of the RGB. Since the isochrone is less slanted towards the right than the actual CMD, this means that the isochrone simulates more metal poor stellar population than the actual cluster. For Figure 24b, although the HB of the isochrone looks like it is aligned with the CMD, the RGB is not aligned correctly and it looks like it is too metal poor compared to the CMD.

### 3 The Second Parameter Problem - [Extension]

The second parameter problem typically arises in the Horizontal Branch (HB) of globular cluster. With reasonable theoretical argument and observational records, the HB morphology was believed to be influenced by the metallicity of the globular cluster. Using the same explanation to the relationship between the RGB and metallicity, the HB will be more red-shifted if a cluster has higher metallicity. However, some globular clusters have shown the same metallicities but yet their HB morphology differ quite distinctively. This is an evidence that there may be a second parameter which may be a determining factor of HB morphologies.

We have seen the second parameter effect in Figure 18. The younger isochrone has a more red-shifted HB compared to the older one of same metallicity, hence the second parameter in this case is the age. Aaron Dotter's 2011 [paper](#) has experimentally proved that the HB is indeed dependent on age. Figure 25 on the next page shows a result based on 68 globular clusters, adapted from Dotter's paper. HB Type on the x-axis is based on the formula:

$$\frac{B - R}{B + V + R} \quad (2)$$

Where:

$B$  = No. of Blue stars

$V$  = No. of RR Lyrae Variables

$R$  = No. of Red stars

From equation 2, we can deduce that a more positive HB Type implies a bluer morphology. By looking at the plot in Figure 25, there is an obvious age dependence when the metallicity is about at about  $-1.6 < [Fe/H] < -1.3$ . Within this metallicity range, younger clusters (red points) tend to have a redder morphology than older clusters. Which evinced that age is a determining factor of HB morphology but only within this small range of metallicity. For metallicities  $[Fe/H] < -1.6$  and  $[Fe/H] > -1.3$ , the HB morphology is completely independent of age. Here, they become highly dependent on metallicity. This can be observed from Figure 25 where clusters of similar age but different metallicities cover a massive range of HB Type. At these metallicities, clusters of high metallicities have a redder HB morphology as expected.



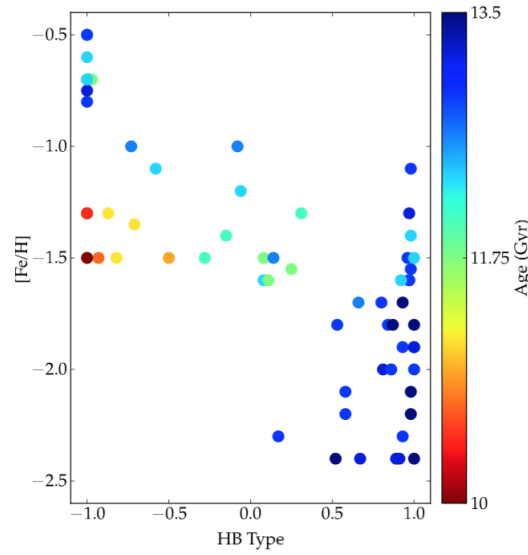


Figure 25: This graph shows the correlation between the HB Type, cluster metallicity and cluster age.

Although the plot shows a clear relationship between HB Type, age and metallicity, there is an observable spread at about 0.5 – 1.0 on the HB type scale. This could be telling us that age and metallicity are not the only influencing factors of the HB morphology. The identity of this extra influencing factor remains a mystery and is currently still a work in progress. Dr. Antonino Milone’s presentation <sup>12</sup> on this topic suggested a few other possible parameters that may be affecting the morphology of the HB, each of which were investigated by different research groups. These parameters include the mass-loss, stellar rotation, planetary systems, magnetic field, helium abundance, globular cluster’s ellipticity, cluster mass, and cluster central density.

<sup>12</sup>The Second Parameter Problem of the Horizontal Branch Morphology in Globular Clusters. Presented by Dr. Antonino Milone of RSAA, ANU.  
Link: <https://www.aao.gov.au/files/palmcove2014/Milone.pdf>

# Distance to Globular Clusters

## 1 Distance Calculation Procedures

To calculate the distance to PA02, the shift values obtained from the isochrone fitting procedure is required. With both the x and y-shifts ( $\Delta x$  and  $\Delta y$ ), we can calculate the distance to the cluster following the equations in the following order:

1. Colour Excess,  $E(V - I)$ :

$$E(V - I) = (V - I)_{\text{observed}} - (V - I)_{\text{intrinsic}} \quad (3)$$

$$E(V - I) = A(V) - A(I) \quad (4)$$

Where:

- $A(X)$  is the extinction in the X-band
- V and I respectively represents the F606W and F814W filters of the ACS/WFC HST used to image PA02.

Color excess is the difference between observed and intrinsic color. Observed color has undergone extinction. Extinction is a phenomenon of light absorption or scattering by interstellar medium such as dust particles and gasses. This causes the observed color to be more red shifted than the actual intrinsic color of the object. Given that reddening happens as a result of extinction, color excess is always positive.

In this case, the isochrone gives the intrinsic color while the CMD (the actual data) has gives the observed color, so we have:

$$\Delta x = E(V - I) \quad (5)$$

2. Total Extinction in the I-band,  $A(I)$ :

$$A(I) = R_I \times E(V - I) \quad (6)$$

The color excess,  $E(V - I)$ , can be used to calculate the extinction in a specific color band as shown in equation 6. With equation 5, it can also be written as:

$$\boxed{A(I) = R_I \times \Delta x} \quad (7)$$

In the case of PA02 taken in the V and I band, we choose to calculate the extinction for the I-band. This is because infrared light has longer wavelength and therefore has a weaker extinction as compared to light in the visible band. To calculate  $A(I)$ , we also need the constant  $R_I$  which represents the average size of dust grains causing extinction in the I-band. This constant is known as the "total-to-selective extinction ratio", which is self-explanatory by looking at equations 4 and 6, where:

$$\boxed{R_I = \frac{A(I)}{A(V) - A(I)}} \quad (8)$$

For simplicity, we neglect the extinctions in M1 and also through any medium that lies between M31 and Milky Way. Hence the value used here is  $R_I = 2.62$ , and it only takes into account any extinction caused by obstructions within the Milky Way.

### 3. Distance Modulus, $m - M$ :

$$\boxed{m - M = 5 \log_{10} \left( \frac{r}{10} \right)} \quad (9)$$

Where:

- $m$  is the apparent magnitude
- $M$  is the absolute magnitude
- $r$  is the distance to object in parsecs (pc)

Distance Modulus is the difference between the apparent and absolute magnitude. The apparent magnitude in our isochrone fits comes from the actual CMD while the absolute magnitude comes from the isochrone. As previously mentioned, electromagnetic radiation (i.e. light) undergoes extinction. Therefore the apparent magnitude intrinsically includes extinction while the absolute magnitude does not. Given that, the Distance Modulus can also be represented by:

$$\boxed{m - M = \Delta y - A(I)} \quad (10)$$

Substituting equation 10 into equation 9 and rearranging it, we get a formula for calculating the distance to the object,  $r$  (i.e. distance PA02 in this case):

$$r = 10 \times 10^{\frac{\Delta y - A(I)}{5}} \quad (11)$$

The procedure above shows notations and constant values specified for the PA02 cluster. As for G1, the procedure is exactly the same except that the V-band is replaced by the B-band and the I-band is replaced by the V-band. This is because G1 was taken in the F438W (B) and F606W (V) filters using the WFC3 as mentioned before. Also, the total-to-selective extinction ratio should now be  $R_V$  rather than  $R_I$  by analogy. So if we once again ignore any extinction beyond the Milky Way, the value of  $R_V$  is 3.19.

## 2 Distance to PA02 - Isochrone Fitting 1

Approximately 25 isochrones were used for the fitting, but only two of the isochrones which best fits the CMD of PA02 were selected for analysis. The first isochrone used is shown in Figure 26 below. For this isochrone, I focused more on fitting the HB rather than the RGB, although the RGB still fits quite well onto the CMD.

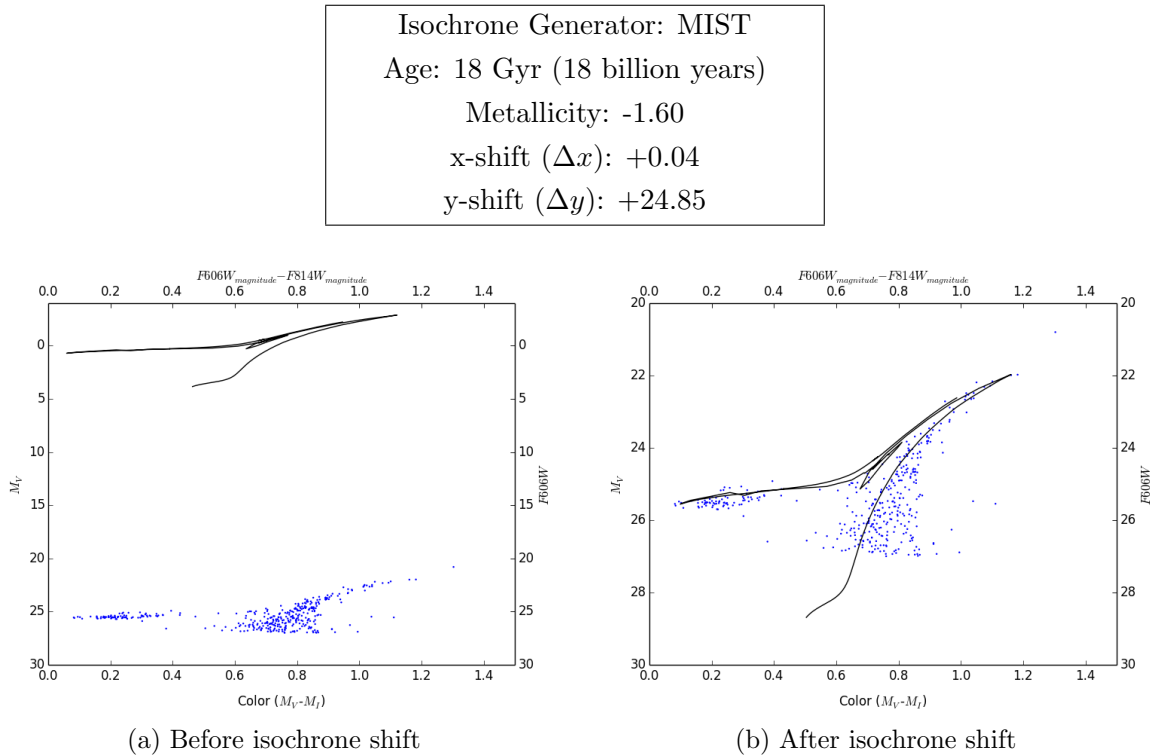


Figure 26: PA02 Fitting - Isochrone 1

### PA02 Isochrone 1 - Distance calculation procedure:

- **Step 1: Extinction,  $A(I)$**

$$\begin{aligned} A(I) &= 2.62 \times \Delta x \\ &= 2.62 \times (+0.04) \\ &= 0.1048 \end{aligned}$$

- **Step 2: Distance of PA02**

$$\begin{aligned} r &= 10 \times 10^{\frac{\Delta y - A(I)}{5}} \\ &= 10 \times 10^{\frac{24.85 - 0.262}{5}} \\ &= 8.893 \times 10^5 pc \\ &\approx 890 kpc \end{aligned}$$

#### Comments:

According to Huxor et al.<sup>13</sup> the projected galactocentric radius ( $R_{proj}$ ) of PA02 is 114.74 kpc.  $R_{proj}$  is the distance of an object from a galaxy's galactic core as seen on a projected 2-D plane. Hence a non-zero  $R_{proj}$  can imply that an object lies either to the fore or at the rear of the galactic plane. Since the proximity of M31 from earth is approximately 780 kpc, the calculated distance 890 kpc is a rather reasonable value and it suggests that PA02 lies behind the galactic plane of M31.

## 3 Distance to PA02 - Isochrone Fitting 2

The second selected isochrone fitting was generated using the Padova generator. In contrast to the fitting with Isochrone 1, this fitting prioritizes the alignment of the isochrone RGB onto the CMD RGB rather than the HB. The isochrone before and after fitting is shown in Figure 27 on the next page.

Isochrone Generator: Padova  
Age: 13.5 Gyr (13.5 billion years)  
Metallicity: -1.40  
x-shift ( $\Delta x$ ): +0.10  
y-shift ( $\Delta y$ ): +24.43

<sup>13</sup>The outer halo globular cluster system of M31 - I. The final PAndAS catalogue. Huxor, Avon, D. Mackey, A and M. N. Ferguson et al. (2014).

**Link:** <https://arxiv.org/pdf/1404.5807.pdf>

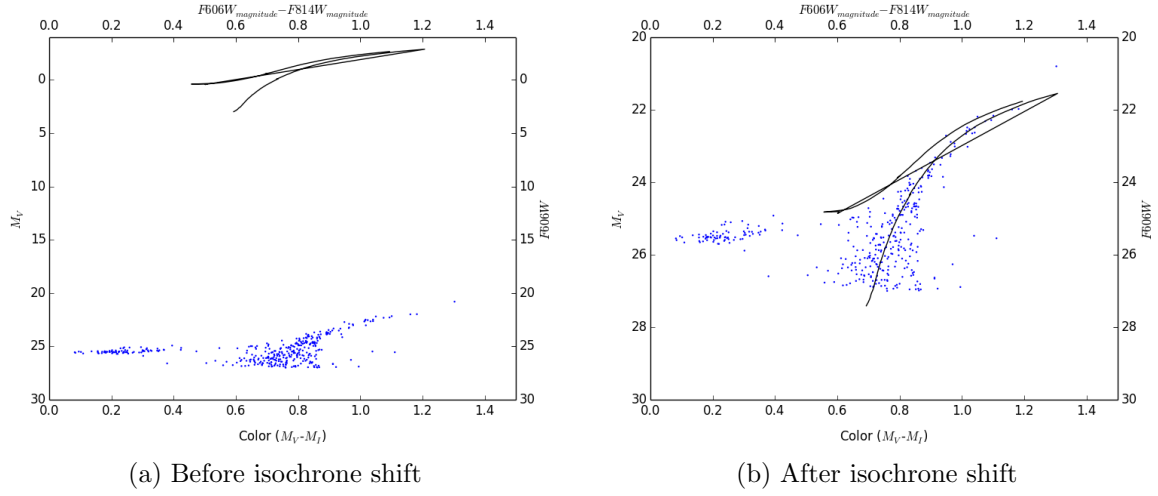


Figure 27: PA02 Fitting- Isochrone 2

**PA02 Isochrone 2 - Distance calculation procedure:**

- **Step 1: Extinction,  $A(I)$**

$$\begin{aligned}
 A(I) &= 2.62 \times \Delta x \\
 &= 2.62 \times (+0.10) \\
 &= 0.262
 \end{aligned}$$

- **Step 2: Distance of PA02**

$$\begin{aligned}
 r &= 10 \times 10^{\frac{\Delta y - A(I)}{5}} \\
 &= 10 \times 10^{\frac{24.43 - 0.262}{5}} \\
 &= 6.817 \times 10^5 pc \\
 &\approx 682 kpc
 \end{aligned}$$

Comments:

From this isochrone fit, the calculated distance is 682 kpc which is still a reasonable value and is within range of  $R_{proj}$ . However, this distance suggests that PA02 lies in the foreground of the galactic plane of M31 which contradicts the results of the first isochrone fitting.

Conclusion:

One may say that the second isochrone fitting is less accurate since the isochrone HB does not resemble that of the CMD. However, there is a suspicion in the first isochrone where the age is 18 Gyr. Our universe is only about 13.8 Gyr old, so it is impossible for a cluster to form before the existence of the universe. The 18 Gyr isochrone could just be a theoretical prediction of how

stellar clusters would be like in the future, or it could also be an error due to the difficulty in converting computational time-step into actual time-step. The reason is the existence of an 18 Gyr isochrone is unknown and hence I cannot conclude the exact distance of PA02 from earth based on the two isochrone fittings shown above. The only inference I can make from looking at these results is that PA02 lies in the distance range of approximately 650 – 980 kpc from earth, with an estimated uncertainty of  $\pm 30$  kpc.

## 4 Distance to G1 - Isochrone Fitting 1

Similar to PA02, two isochrone fittings were also selected for the G1 cluster to determine its distance. The first isochrone is once again selected based on the best HB alignment and is shown in Figure 28. Recall that the G1 cluster has a spread of metallicity, hence two isochrones were used to fit each of the G1 CMD rather than single isochrone fittings like PA02. Each of the two isochrones represent the upper and lower bound metallicity of the cluster respectively. That being the case, aligning the isochrones' RGB and HB onto the CMD with the same x and y-shifts is rather difficult, making the fitting process more tedious than PA02.

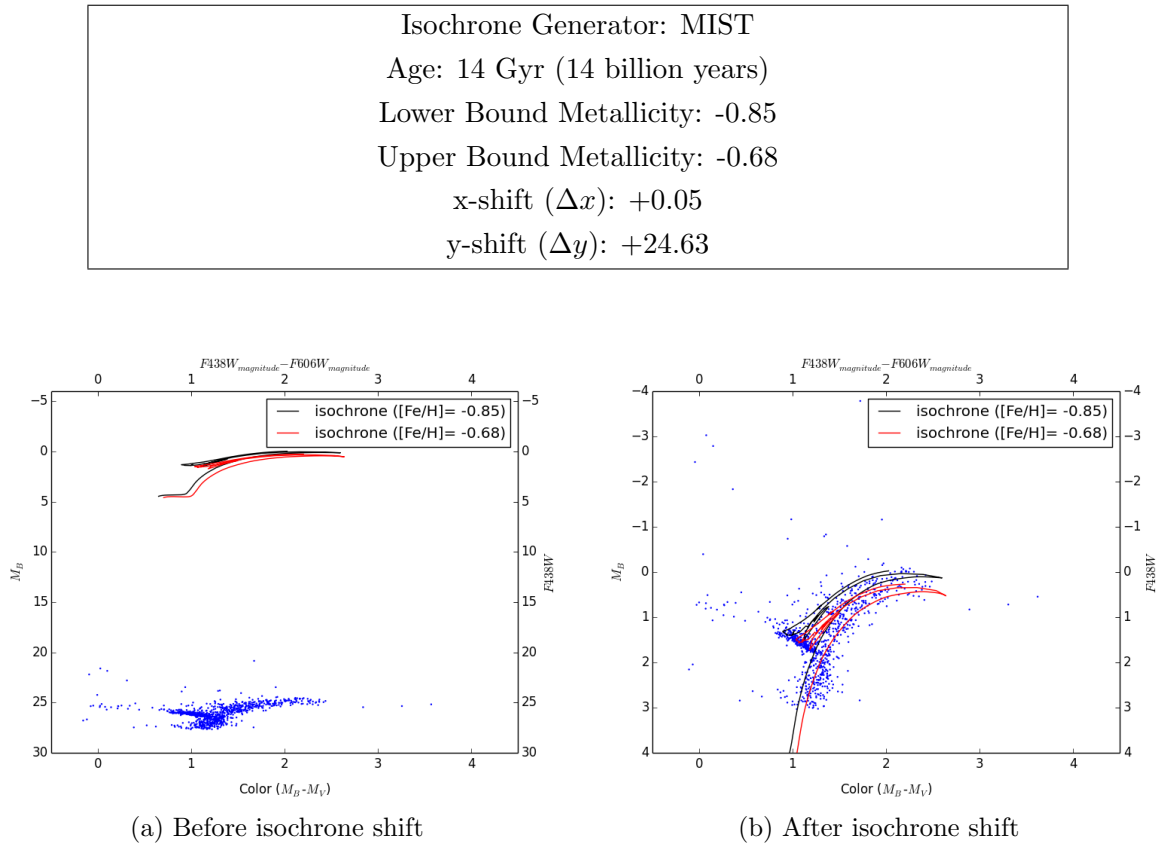


Figure 28: G1 Fitting- Isochrone 1

### G1 Isochrone 1 - Distance calculation procedure:

- **Step 1: Extinction,  $A(V)$**

$$\begin{aligned} A(V) &= 3.19 \times \Delta x \\ &= 3.19 \times (+0.05) \\ &= 0.1595 \end{aligned}$$

- **Step 2: Distance of G1**

$$\begin{aligned} r &= 10 \times 10^{\frac{\Delta y - A(V)}{5}} \\ &= 10 \times 10^{\frac{24.63 - 0.1595}{5}} \\ &= 7.836 \times 10^5 pc \\ &\approx 784 kpc \end{aligned}$$

#### Comments:

According to Meylan et al.<sup>14</sup>, the  $R_{proj}$  of G1 is about 40 kpc. Therefore, the results with G1 lying at 784 kpc away from earth tells us that it is located at about about 4 kpc behind the galactic plane of M31.

## 5 Distance to G1 - Isochrone Fitting 2

The second set of isochrones were also generated by MIST but they are of different metallicity ranges as compared to the first set of isochrones. This fitting prioritizes the alignment of the RGB rather than the HB. I personally think that this is the best RGB fitting for G1, but not everyone would agree with me. As previously mentioned, to get a good isochrone fitting with the "fitting-by-eye" technique is highly arbitrary. It solely depends on how one defines a good fitting. In the case of G1, I defined a good RGB fitting as: when the two of the selected isochrones encapsulates the most number of the stars at the TRGB.

|                                |
|--------------------------------|
| Isochrone Generator: MIST      |
| Age: 12 Gyr (12 billion years) |
| Lower Bound Metallicity: -0.68 |
| Upper Bound Metallicity: -0.53 |
| x-shift ( $\Delta x$ ): +0.08  |
| y-shift ( $\Delta y$ ): +24.50 |

<sup>14</sup>MAYALL II G1 IN M31 : GIANT GLOBULAR CLUSTER OR CORE OF A DWARF ELLIPTICAL GALAXY? Meylan et al. (2001)

Link: <http://iopscience.iop.org/article/10.1086/321166/pdf>



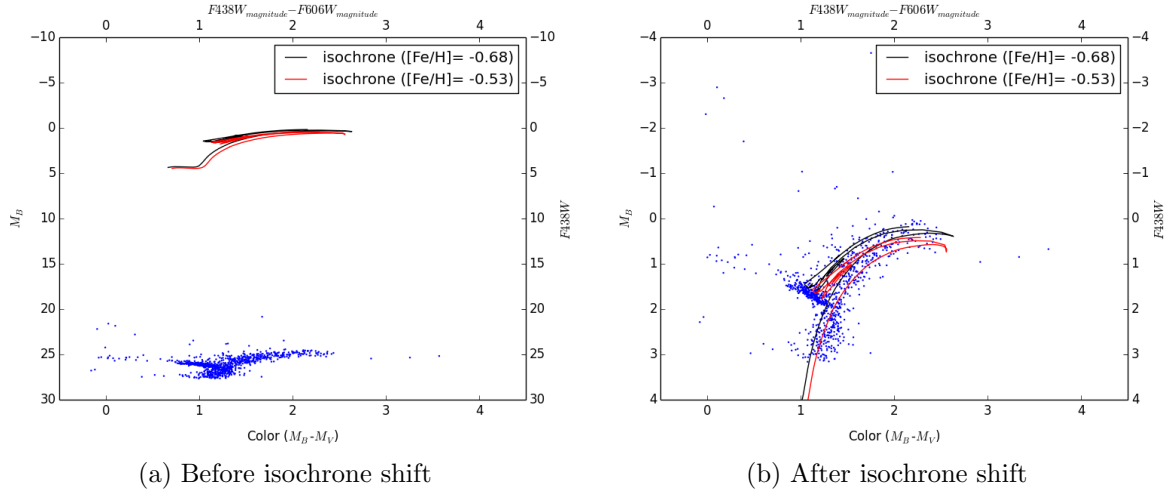


Figure 29: **G1 Fitting- Isochrone 2**

### G1 Isochrone 2 - Distance calculation procedure:

- **Step 1: Extinction,  $A(V)$**

$$\begin{aligned}
 A(V) &= 3.19 \times \Delta x \\
 &= 3.19 \times (+0.08) \\
 &= 0.2552
 \end{aligned}$$

- **Step 2: Distance of G1**

$$\begin{aligned}
 r &= 10 \times 10^{\frac{\Delta y - A(V)}{5}} \\
 &= 10 \times 10^{\frac{24.50 - 0.2552}{5}} \\
 &= 7.063 \times 10^5 pc \\
 &\approx 706 kpc
 \end{aligned}$$

### Comments:

The calculated distance being 706 kpc with M31 lying at 780 kpc may seem like it is out of the  $R_{proj}$  range. However, that is not necessarily the case as G1 may lie 40 kpc away from the galactic centre when projected on a 2-D plane, but it can be lying at a radial distance of more than 40 kpc in the fore of M31. Hence the result is still logically acceptable but it once again contradicts the results of the last isochrone fitting where G1 was said to lie further behind M31's plane.

### Conclusion:

The distances of G1 calculated using two different sets of isochrones also do not agree for the G1 cluster. Therefore we once again cannot conclude the distance of G1 but we can say that

it lies somewhere between 670 – 820 kpc with an estimated error range of 30 kpc as per PA02. With the both the fittings of PA02 and G1, we can speculate that isochrone fitting by-eye could sometimes provide ambiguous and unreliable results, yet the process tedious and time consuming.

## **6 Improving the Results of Isochrone Fitting**

Of course, to use a computational method for isochrone fitting would be the most ideal solution for improvement. However, like we said before, there is yet to be a non-controversial method for conducting isochrone fitting very accurately. Therefore we need to look into things that could be improved for the isochrone fitting "by-eye" method.

A way the results could be improved is to select isochrones in a more thorough manner. For the isochrone fittings in this project, only the age and metallicity was taken into account when generating the isochrones for the fittings. By focussing on only two parameters, the isochrone may have neglected some other parameters and it could affect the accuracy of the fitting. Some possible parameters to consider, for future reference, in order to obtain potentially better results include: helium abundance, alpha element abundance ( $[\alpha/Fe]$ ), and star cluster age range. The age range problem applies more towards G1 as it was thought to have developed over a long time scale. Thus the stars in the cluster may be of different ages where using single-aged isochrones for the G1 fitting might have been a trivial process, although we are still not entirely sure about the reason for the result inaccuracy.

## The Gemini North Telescope

We now move on to the final section of the project, which is to experiment on running DOLPHOT photometry for ground-based telescope data. The telescope from which the data were collected is the Gemini North Telescope (see Figure 30) located on the summit of Mauna Kea, on the island of Hawaii – one of the best sites for astronomical observations in the world.



Figure 30: The Frederick C. Gillett Gemini North telescope. (Adapted from [Gemini](#))

This 8.19-meter telescope is equipped with adaptive optics<sup>15</sup> to correct for atmospheric distortion and hence it provides one of the highest quality ground-based imaging in the world. However, comparing the image resolution to that of the HST, light reaching Gemini is still subject to atmospheric turbulence effects which may not be perfectly corrected by the adaptive optics system. This will then affect the image sharpness. On top of that, the adaptive optics system is only capable of correcting atmospheric distortion. It cannot retrieve the light which has been absorbed by Earth's atmosphere before it reaches the telescope. Therefore, although space telescopes cost a lot more and are much harder to maintain, they definitely have a much higher advantage against ground-based telescopes.

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<sup>15</sup>Telescopes and Sites - Gemini Observatory. Brambold 2015.

**Link:** <http://www.gemini.edu/sciops/telescopes-and-sites>

# DOLPHOT and Ground-based Data

## 1 Aim of the Experiment

It was previously mentioned that DOLPHOT is specially designed to carry out photometry on the HST data. However, an additional function of DOLPHOT compared to HSTphot is that DOLPHOT has a generalized module which can be used to perform photometry on any dataset. This experiment will attempt to make use of this general module to run DOLPHOT on a few globular clusters images taken using the Gemini Telescope. This has never been done before and hence the purpose of this experiment is to test if DOLPHOT could produce decent photometry for ground-based data.

It is logical to think that running DOLPHOT on ground-based data would not work because of the very crowded images with reduced sharpness due to atmospheric disturbances. However, if DOLPHOT can successfully fit the model PSFs onto these ground-based images to produce decent results, it will be very helpful to astronomers who will be observing more M31 globular clusters in the future. This is because ground-based telescopes are more accessible as compared to the HST. The HST is very high in demand due to its excellent imaging ability. Hence, a very solid scientific reasoning with significant research contribution is required to obtain observing time for the telescope. Since observing globular clusters in M31 does not have a sufficiently great scientific contribution, it is almost impossible to get observing opportunities for globular clusters on the HST unless it is within range of another target that is being imaged. Ground-based telescopes are then only options left.

## 2 DOLPHOT Procedure for Gemini data

The procedure is very similar to the HST procedure, except that a header file is required and the "applymask" step is eliminated since the mask image for the Gemini data is unavailable. The "splitgroup" step is also not required here as our images are already in the single-chip format.

1. Prepare **FITS header file** in the working directory to tell DOLPHOT the meaning of headers used in the Gemini ".fits" datasets (see Figure 31).

```
# Header keywords
gain_kw = GAIN
gain_val = 5.0
exptime_kw = EXPTIME
exptime_val = 1.0
rdnoise_kw = RDNOISE
rdnoise_val = 7.155
minval_kw =
minval_val = -100.0
maxval_kw =
maxval_val = 110000.0
mjd_kw = MJD-OBS
mjd_val = 56901.454
airmass_kw = AIRMASS
airmass_val = 1.186
```

Figure 31: Header file for Gemini data.

2. Prepare **text file in the working directory containing parameters** suitable for the general DOLPHOT module. The parameter file used is a revised version of Blair Conn's original parameter file used for the Gemini South dataset. It includes a set of image-specific (Figure 32) and photometry-specific parameters (Figure 33).
3. **Run "calcsky"** on all .fits file.
4. **Run DOLPHOT**

#### Suggested *Terminal* command lines:

- **Run "calcsky":**

~/path/to/dolphot2.0/bin/calcsky (filename) 25 40 -256 2.25 2.25

\*Repeat for all files

- **Run DOLPHOT:**

~/path/to/dolphot2.0/bin/dolphot (cluster-name).phot -pGemini.param

*\*NB: There is no need to specify the header file as DOLPHOT will automatically search for it in the working directory*

```

# Image-specific parameters
Nimg = 2
photsec = 1 1 670 0 5370 4200
img0_file = HEC6_i
img0_psfa = 3.8 0 0 0 0 0
img0_psfb = 3.8 0 0 0 0 0
img0_psfc = 0.0 0 0 0 0 0
img1_file = HEC6_g
img1_shift = 0.0 0.0
img1_xform = 1 0 0
img1_psfa = 5.0 0 0 0 0 0
img1_psfb = 5.0 0 0 0 0 0
img1_psfc = 0.0 0 0 0 0 0
img1_RAper = 3.0
img1_RSky = 12.0 30.0
img1_aprad = 10.0
img1_apsky = 12.0 30.0
img2_file = HEC6_i
img2_shift = 0.0 0.0
img2_xform = 1 0 0
img2_psfa = 3.8 0 0 0 0 0
img2_psfb = 3.8 0 0 0 0 0
img2_psfc = 0.0 0 0 0 0 0
img2_RAper = 3.0
img2_RSky = 12.0 30.0
img2_aprad = 10.0
img2_apsky = 12.0 30.0

```

Figure 32: **Example of Image-specific parameters for the HEC6 cluster.** Notice that only two images were used for running DOLPHOT on Gemini data as each of the image used is already stacked using three separate images. Also, the I-band image (HEC6\_i) of shorter wavelength is used as the reference image rather than the G-band image because light of shorter wavelength experiences less atmospheric distortion and absorption.

```

# Photometry-specific parameters
RChi = 2.5          #centroiding aperture size (flt)
SearchMode = 1      #use chi only for centroid (int 0=no,1=yes)
UseWCS = 2          #use WCS info for alignment?
PSFPhot = 1         #photometry type (int/0=aper,1=psf,2=wtd-psf)
PSFPhotIt = 2       #PSF-fitting iterations (int)
FitSky = 1          #fit sky? (int/0=no,1=yes,2=small,3=with-phot)
RSky0 = 12.0        #inner sky radius (flt>=RAper+0.5)
Rsky1 = 30.0        #outer sky radius (flt>=RSky0+1)
SkipSky = 1         #spacing for sky measurement (int>0)
SkySig = 2.25       #sigma clipping for sky (flt>=1)
NegSky = 1          #allow negative sky values during phot? (0=no,1=yes)
SecondPass = 3      #second pass finding stars (int 0=none,>0=yes)
SigFind = 2.5       #sigma detection threshold (flt)
SigFindMult = 0.85  #Multiple for quick-and-dirty photometry (flt>0)
SigFinal = 3.5      #sigma output threshold (flt)
MaxIT = 25          #maximum iterations (int>0)
NoiseMult = 0.05    #noise multiple in imgadd (flt)
FSat = 0.999        #fraction of saturate limit (flt)
Align = 3           #align images? (int 0=no,1=const,2=lin,3=cube)
AlignIter = 1       #number of alignment iterations
AlignTol = 2        #alignment tolerance
AlignStep = 1       #stepsize for search in AlignTol
Rotate = 1          #allow cross terms in alignment? (int 0=no, 1=yes)
RCentroid = 2       #centroid box size (int>0)
PosStep = 0.1       #search step for position iterations (flt)
dPosMax = 3.0       #maximum single-step in position iterations (flt)
RCombine = 2.0      #minimum separation for two stars for cleaning (flt)
RPSF = 13           #PSF size (int>0)
SubPixel = 1        #subpixel sampling for PSF generation (int >0)
FPSF = G+L          #PSF type (string)
SigPSF = 10.0       #min S/N for psf parameter fits (flt)
EPSF = 1            #elliptical PSFs (int 0=no,1=yes)
Force1 = 0          #force type 1/2 (stars)? (int 0=no,1=yes)
MinS = 2.0          #min FWHM of good star (flt)
MaxS = 10.0         #max FWHM of good star (flt)
MaxE = 0.5          #max ellipticity of good star (flt)
PSFStep = 0.05      #step size for PSF solution
PSFsol = 1          #Order of PSF solution (int -1=none,0=const,1=lin,2=quad)
PSFres = 1          #Compute PSF residual (int 0=no,1=yes)
ApCor = 1           #Compute aperture corrections (int 0=no,1=yes)
Zero = 30.00        #nominal zero point
VerboseData = 1     #make a log file

```

Figure 33: **Photometry-specific parameters.** Applicable to all clusters imaged using Gemini.

### 3 WCS Alignment

A very important factor which determines the quality of the photometry is the alignment of the images. All the images used in the photometry should be aligned properly so that DOLPHOT can fit the PSF models onto the actual PSFs to give an accurate judgement on whether the object is a good star. Figure 34 compares the CMDs produced from good and bad alignments.

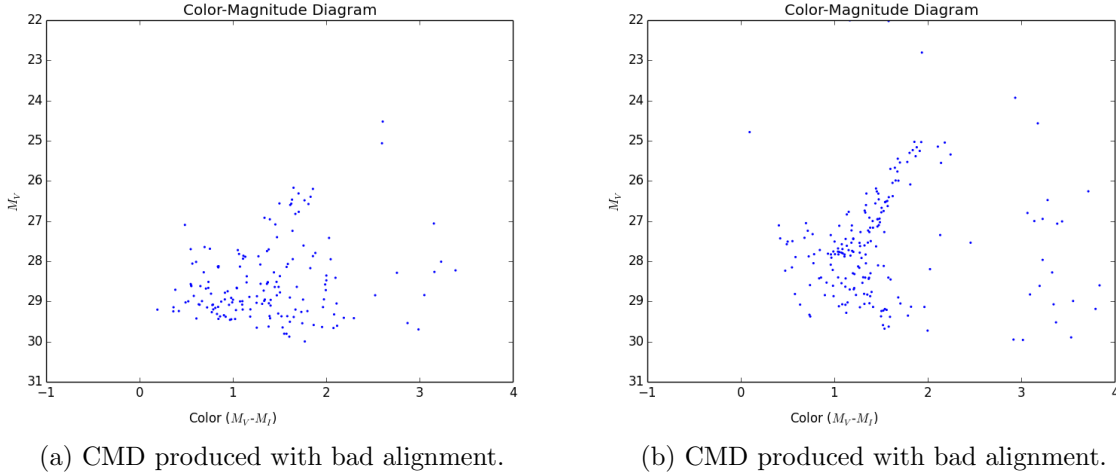


Figure 34: **Good and bad alignment CMD comparison**

The bad alignment was 50 pixels offset in the x-direction while the good alignment was done using WCS header information. It is obvious from the comparison that a 50 pixel offset in alignment can have such a great impact on the CMDs produced. Therefore, we can conclude that an accurate alignment is indeed crucial for producing a good quality photometry.

To use WCS header information as an initial guess for the alignment with a full distortion solution estimated, the parameter file setting should be `UseWCS = 2`. If we prefer to perform manual alignment by comparing the pixel coordinates of the same objects between the images, turn off the WCS alignment by setting `UseWCS = 0`. Then changes should be made to `img_shift` for non-reference images, where the first value is x-axis offset,  $x_{img} - x_{ref}$ , and the second value is the y-axis offset,  $y_{img} - y_{ref}$ . An example of manual alignment is shown below.

```

photsec = 1 1 670 0 5370 4200
img0_file = PA16_i
img0_psfa = 3.8 0 0 0 0 0
img0_psfb = 3.8 0 0 0 0 0
img0_psfrc = 0.0 0 0 0 0 0
img1_file = PA16_g
img1_shift = 169.03 -170.48
img1_xform = 1 0 0
img1_psfa = 5.0 0 0 0 0 0
img1_psfb = 5.0 0 0 0 0 0
img1_psfrc = 0.0 0 0 0 0 0
img1_RAper = 3.0
img1_RSky = 12.0 30.0
img1_aprad = 10.0
img1_apsky = 12.0 30.0

```

Figure 35: Image-specific parameter for manual alignment. (`img0` is the reference image)



## 4 Filtering Gemini DOLPHOT Output

The parameters being filtered in the Gemini DOLPHOT output are analogous to DOLPHOT ACS/WFC and WFC3. There are only two modifications:

1. Recall from Figure 32 that each image used for photometry is already stacked. So the first difference is that the instrumental magnitude cut is now implemented on single image magnitude in the G ( $M_V$ ) and I ( $M_I$ ) band respectively, rather than on the average magnitude of 3 images as per PA02 and G1.
2. Previously for PA02 and G1, the selection radial cut only limits the outer radius. For the clusters imaged by Gemini, an additional cut for the inner selection radius is required to produce a better quality CMD. To show the importance of the additional output filter, CMDs with and without the inner radial cut of a crowded globular cluster, SDSS-G, is shown in Figure 36 below. The image of SDSS-G can be found in Figure 46.

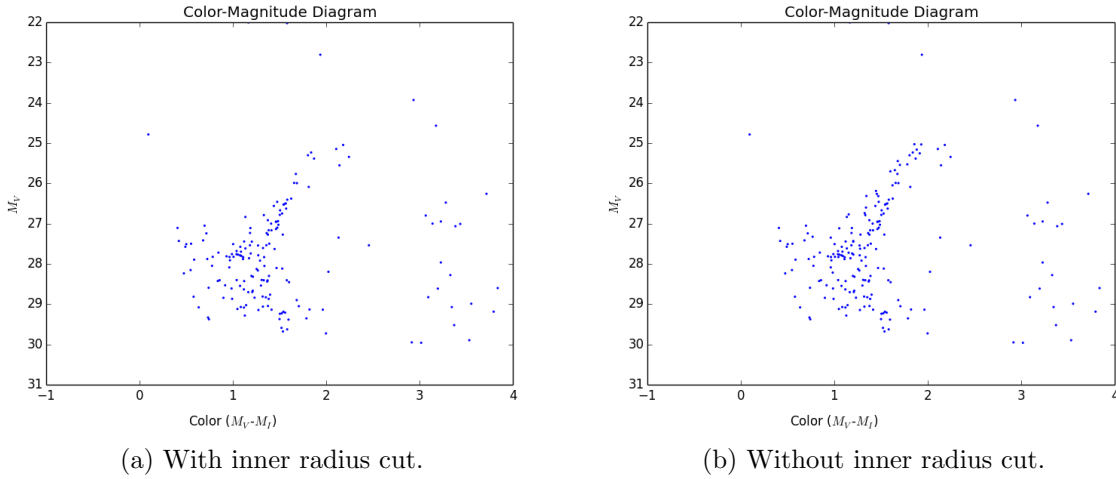
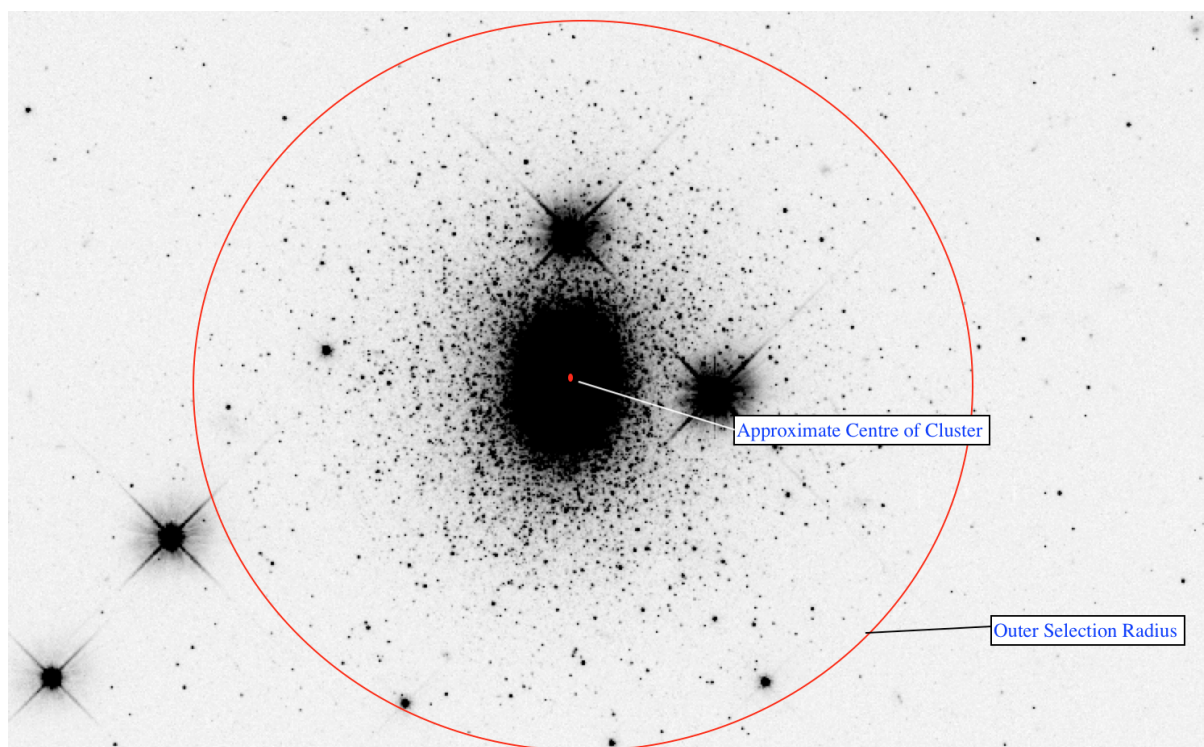


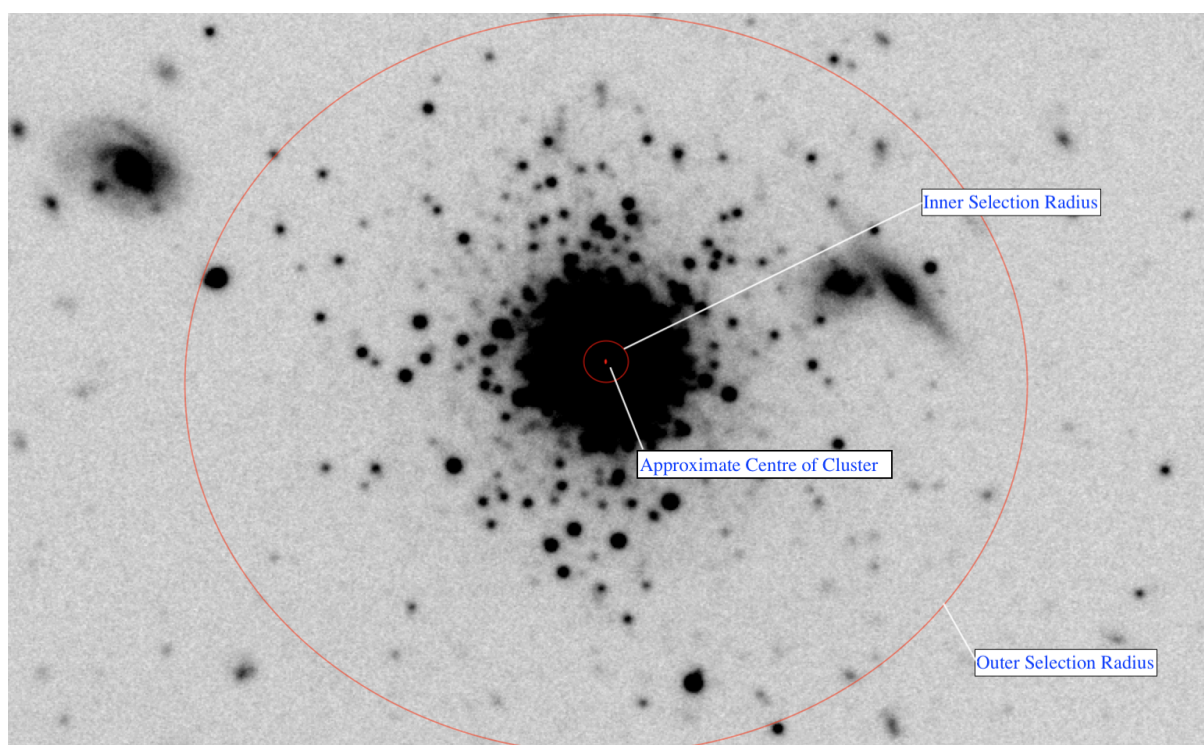
Figure 36

It is very obvious that the CMD of SDSS-G with the inner radius cut has a more well-defined RGB. This is because implementing the inner radius cut eliminated trivial objects that DOLPHOT has found at the centre region of the cluster. Stars in the centre region of highly crowded cluster should not have a defined edge and should have very high crowding, so DOLPHOT is not supposed to be finding any stars at all otherwise it should have been eliminated by the crowding filter. However if DOLPHOT actually managed to find any stars in this region which somehow passed the crowding condition, it may have been due to DOLPHOT trying to fit PSF models onto random points which are not actually stars.

Illustrations of ground-based and the HST photometry radial cuts are shown in Figure 37 on the next page. Note that the circles in the images are only for illustration purposes and they may not be the actual star selection region. The actual selection region should be much larger than what is shown in the illustrations.



(a) Outer selection radius illustrated for the G1 cluster image taken by the HST.



(b) Inner and outer selection radius illustrated for the H25 cluster image taken by the Gemini Telescope.

Figure 37: Comparing the radial selection region for space and ground-based telescopes

# Ground-based Photometry Results

## 1 DOLPHOT Output Filter Conditions

- **Inner Selection Radius:**  $r > 70$  pixels
- **Outer Selection Radius:**  $r < 500$  pixels
- **Object Type:** Type 1 only
- **Colour Uncertainty:**  $C_{Err} < 0.5$

\*Conditions below apply to all X-band (e.g. G and I)

- **Error Flag in X-band:**  $F_{Err_X} \leq 4$
- **Instrumental Magnitude in X-band:**  $M_X < 30$
- **Sharpness in X-band:**  $-0.30 < X_{sharpness} < 0.30$
- **Crowding in X-band:**  $X_{crowding} < 0.35$

Notice that the crowding and sharpness limit is higher than the HST limit which were 0.10 and  $\pm 0.10$  respectively. This is once again due to the higher crowding and reduced sharpness problem in ground-based imaging. Although setting the conditions to be more lenient may let some trivial (non-star) objects pass the filters, but it is done in order to retain the useful stars for plotting the CMD.

\*NB: G-band magnitude is indicated by  $M_V$  and I-band magnitude is indicated by  $M_I$  on the CMD axes.

The globular clusters investigated were: H25, HEC13, HEC6, PA16 and SDSS-G. There clusters can be found in the stellar halo of M31.

## 2 Halo-25 (H25)

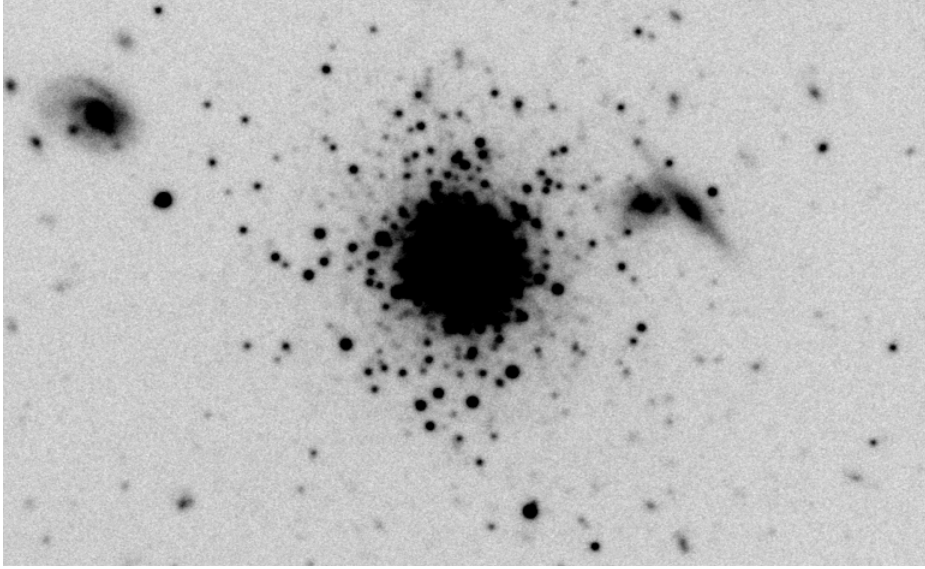


Figure 38: Image of globular cluster H25 taken by Gemini North Telescope.

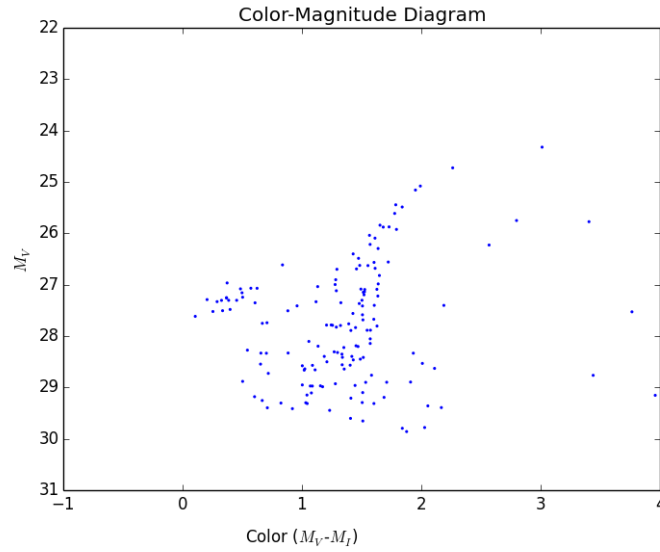


Figure 39: CMD of H25.

### Comments:

In this CMD, we can see the HB and the RGB very clearly where the HB has a blue-ish morphology with maybe a few RR Lyrae variables. However, the CMD is very spread out at high magnitudes and this region may contain a lot of "junk" due to the less stringent conditions imposed on the Gemini DOLPHOT output. We have yet to figure out how to omit them without discarding useful stars.

### 3 Halo extended cluster 6 (HEC6)

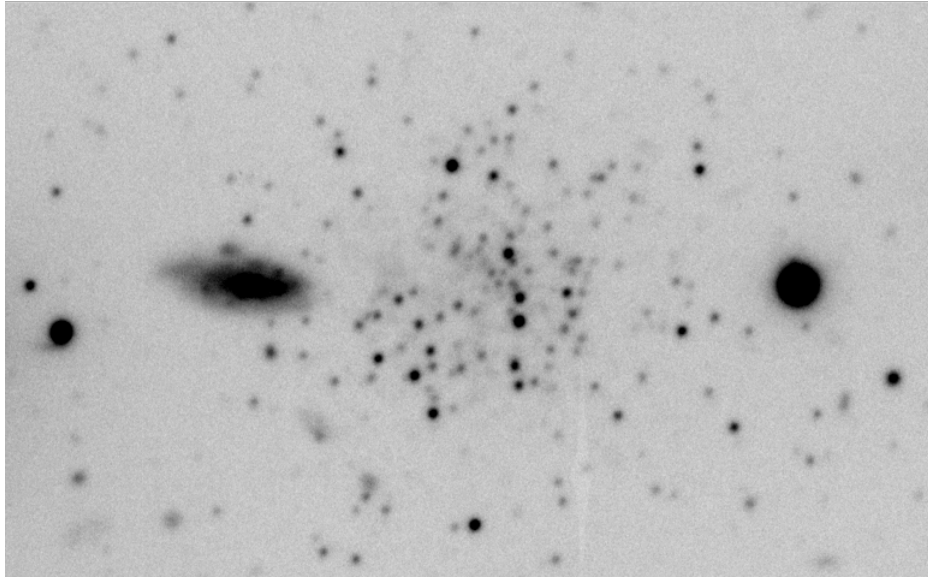


Figure 40: Image of globular cluster HEC6 taken by Gemini North Telescope.

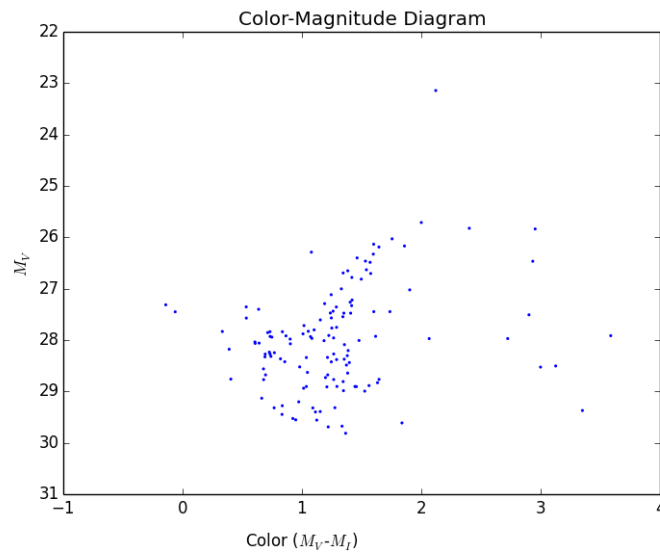


Figure 41: CMD of HEC6.

#### **Comments:**

The CMD is less spread out at fainter magnitudes in this case, but it looks like there is a lot of objects in the redder end of the color scale that might not be part of the cluster. Also, a less obvious HB with a rather red morphology and possibly containing a few blue stars can be seen at magnitudes of about 27-29.



#### 4 Halo extended cluster 13 (HEC13)

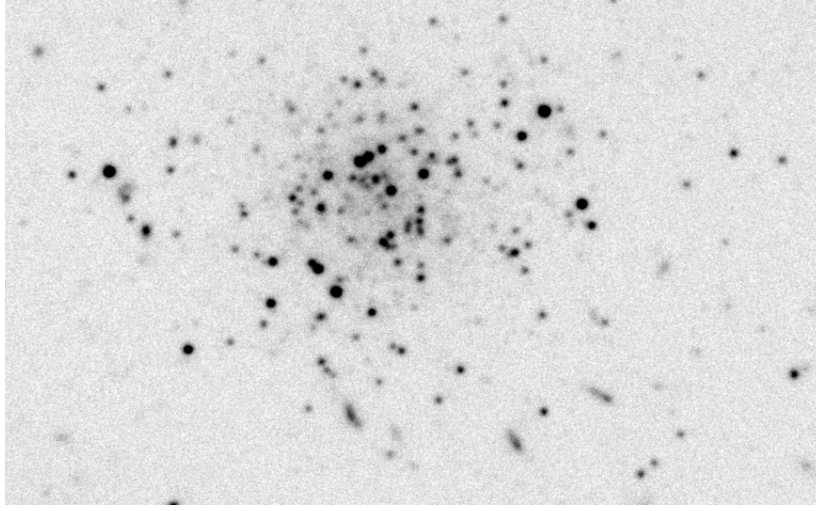


Figure 42: Image of globular cluster HEC13 taken by Gemini North Telescope.

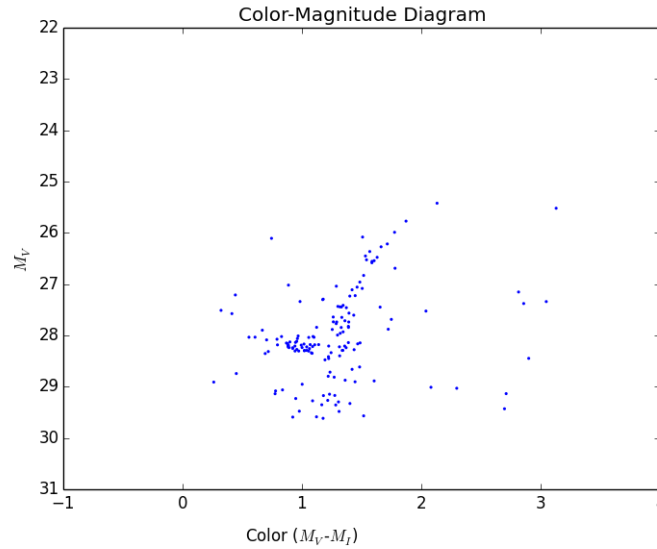


Figure 43: CMD of HEC13.

##### **Comments:**

We can still see a few random objects (the "junks") in the redder end of the color scale which we failed to remove. However, compared to the H25 and HEC6 CMDs, this CMD is not as spread out at fainter magnitudes. Also, it has a much more well-defined RGB and HB. The HB in this cluster has a red morphology with almost no blue stars. This implies that the photometry done for this cluster was of good quality which could probably be due to the cluster having stars that are more spread out. This means that there will be less crowding where DOLPHOT can easily detect objects which are not stars and these objects can be easily removed by the filter without worrying about losing useful stars.

## 5 PAndAS 16 (PA16)

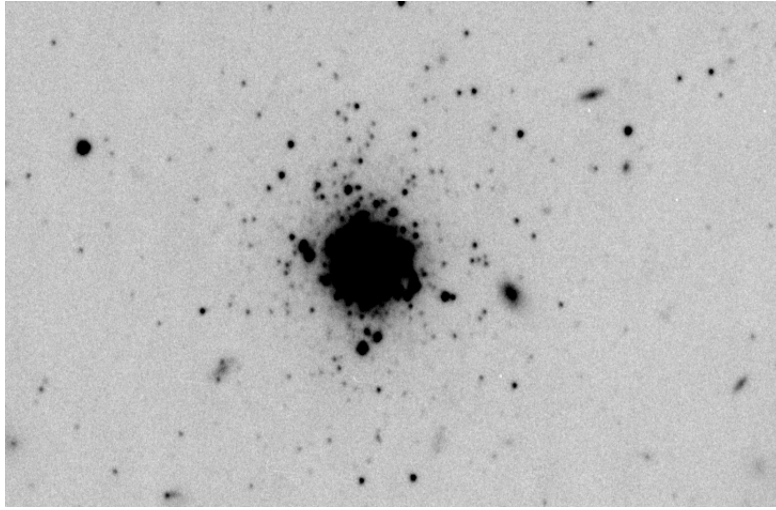


Figure 44: Image of globular cluster PA16 taken by Gemini North Telescope.

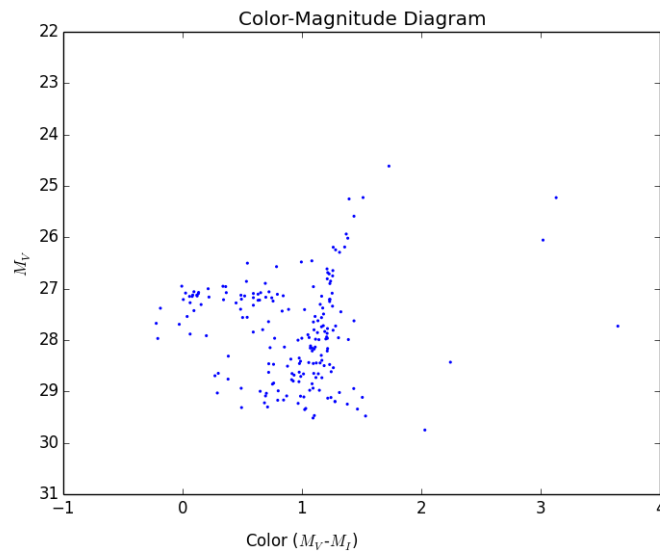


Figure 45: CMD of PA16.

### **Comments:**

For this cluster, the original parameter with WCS alignment did not work due to some error called "illegal cross term" which we have no idea what it means. However, the photometry worked perfectly fine with manual alignment. The CMD of this cluster has minimal unwanted objects. On top of that, it also has an obvious RGB and a nice HB extending across a relatively wide range of color, although favours more towards the bluer end on the color scale. The HB may even contain a few RR Lyrae variables. This could be telling us that good manual alignment may result in a better quality photometry than the WCS alignment. However, we are still unsure if that is the case, so further experiments have to be done to confirm this.

## 6 Sloan Digital Sky Survey: SDSS-G

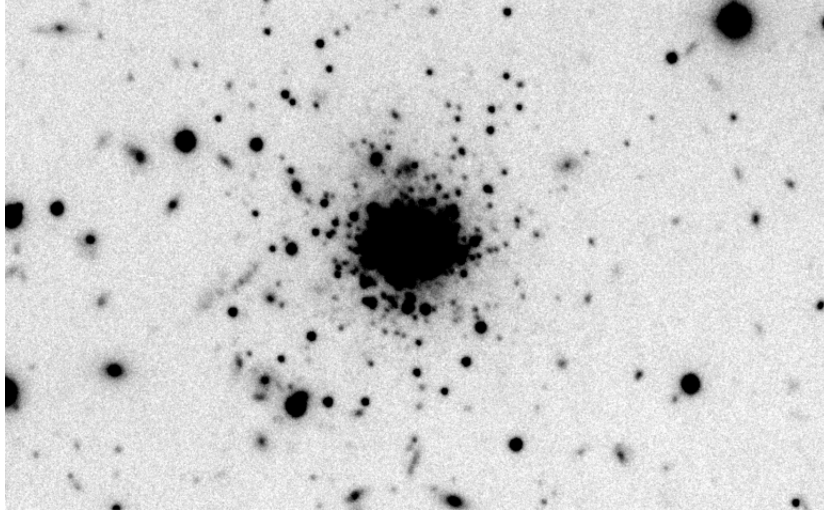


Figure 46: Image of globular cluster SDSS-G observed by the Sloan Digital Sky Survey (SDSS).

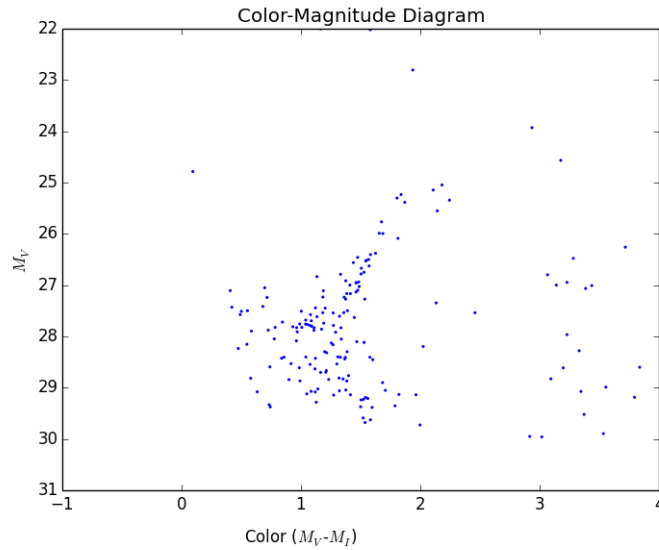


Figure 47: CMD of SDSS-G.

### Comments:

Similar to the CMD of HEC6 and HEC13, there is a lot of random objects in the redder end of the color scale. It also has an analogous HB to HEC13 as it has a relatively red morphology. However, the tip of the RGB is not as well defined as the other 4 clusters shown previously. Perhaps this could be telling us that the cluster has a metallicity spread just like G1? Again, we have no evidence to justify this guessed reasoning so further investigation has to be done to understand this occurrence.



## 7 Conclusion: Ground-Based Photometry with DOLPHOT

From all the five CMDs observed for globular clusters H25, HEC6, HEC13, PA16 and SDSS-G, it seems like DOLPHOT can work decently well with ground-based data even though the quality may not be of the same standard as the HST data. Due to time constraints, I did not get to analyze the sharpness and crowding plots in detail nor did I manage to fit any isochrones for these clusters. Nevertheless, the fact that using DOLPHOT photometry on ground-based data can produce reasonably good CMDs is definitely a promising result and this could be of great help to future research work concerning globular clusters of M31.

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